# Measuring the Task Value of Precision Upgrades in Closed-Loop Agents

Anonymous ACL submission

## Abstract

How much does a precision upgrade help a low-precision agent complete a task? We measure three interventions at a restored decision state: replacing the current action with BF16's, taking over the continuation, or both. Audited replay enables paired comparisons of rescues and damages. On Qwen3 agents under simulated quantization in $\tau^2$-bench, replacing the first tool-structure-divergent W4 action changes 15.5–20.9% of outcomes, with pooled net gain $+0.7$ points (new-task 95% interval $[-2.1, +3.4]$). The pre-registered telecom holdout estimate is negative ($-4.6$ points; 90% $[-7.2, -2.0]$); the protocol lacked a negative-effect category and classifies it as inconclusive. A second W4 checkpoint also produces two-sided outcome changes. At common states, BF16's extra action value over this checkpoint remains unresolved; equivalence is not established. Reordering tool-argument keys also changes outcomes despite identical tool operations. Finally, a frozen uncertainty trigger enriches outcome-changing decisions without a reliable net gain. These results distinguish sensitivity from the value of a specific upgrade: evaluations need reference interventions and selectors validated by net task benefit.

## 1 Introduction

A low-precision LLM agent (Q) can retain its BF16 reference (B) as a fallback (Li et al., 2026c). Deciding when to use that fallback requires knowing the task benefit of a specific upgrade: B may **replace the current action, take over from the next step, or do both**. These choices intervene at different points in a closed-loop interaction. Average success rates compare complete agents, while KL and action disagreement describe local drift; neither directly measures what a particular fallback decision gains or loses.

Figure 1 shows one decision from our data: Q looks up a reservation id that does not exist, B looks up the user's profile, and of the four combinations {B's or Q's action} $\times$ {B or Q continuing} at the frozen state only BB succeeds. B's action is worth a success only if B also continues, which no evaluation of the action alone can show. The case illustrates the measurement distinction; it is not evidence for a mechanism.

When a forked trajectory is continued with another model (Gonuguntla, 2026), that model writes both the next action and everything after it, so the two effects cannot be separated. We cross the source of the current action with the model that executes the rest of the episode, at a frozen state whose exact replay we audit (§3.3), and read the results against two reference interventions: a second W4 checkpoint of the same model (Q′) in place of B, and an argument key-order change that leaves the tool operation identical. We study Qwen3-4B and Qwen3-8B (Qwen Team, 2025) under simulated weight and KV-cache quantization in $\tau^2$-bench (Yao et al., 2024; Barres et al., 2025); BFCL multi-turn (Patil et al., 2025) and ALFWorld (Shridhar et al., 2021) (other environments) and Mistral-Small-3.2-24B (Mistral AI, 2025) (a second model family, forked at sampled $\tau^2$-bench states) are scope checks (Table 1).

**Contributions.** We develop an auditable protocol for measuring three precision interventions on shared states. Building on step values under a continuing policy (Setlur et al., 2024), the protocol combines action–continuation contrasts, paired rescue–damage accounting and two reference substitutions. Our empirical analysis separates a decision's sensitivity to substitution from the additional task value of higher precision.

The experiments answer three questions:

1. **What does an upgrade change?** Replacing the current W4 action produces substantial rescues and damages, which largely cancel in the average. The telecom holdout has a negative

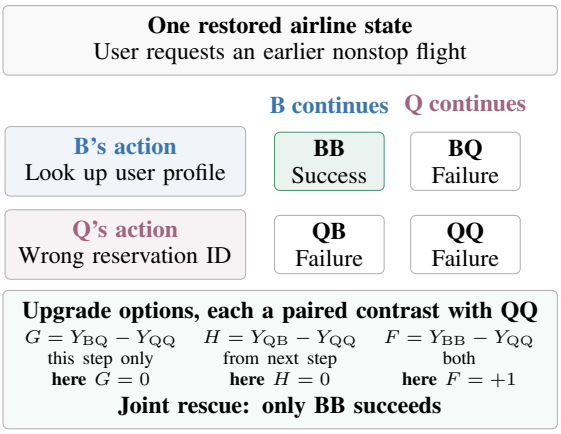



Figure 1: Four branches at one restored state (Qwen3-4B, AWQ-style W4; airline task 31, decision 1 of 12). Only the joint replacement succeeds; action-only and continuation-only evaluations miss this rescue. The case illustrates the measurement distinction, not a mechanism.

estimate; its frozen protocol nevertheless classifies it as inconclusive (§4).

2. **Do flips indicate a need for more bits?** A second W4 checkpoint and a semantically identical key-order change also alter outcomes. A direct common-state comparison does not establish either BF16's extra action value or equivalence (§5).

3. **Can sensitivity guide selection?** One frozen Q-only rule finds more outcome-changing decisions than random selection, but no tested rate reliably improves net gain after correction (§6).

## 2 Related Work

**Quantized agents compared episode by episode.** *Flat Score, Amplified Failures* (Jang et al., 2026) is closest in setting: on $\tau^2$-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, paired tasks flip in both directions, and recovery is equally frequent at both precisions. Its episodes are sampled independently, so the precision of the failing action and that of the recovering controller change together; we vary the two separately at a fixed state. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

**Forks and counterfactual replay.** *The Replay Gap* (Gonuguntla, 2026) forks live SWE-bench Verified trajectories for per-step model switching; its swap replaces the next action and the continuation together (our "both" contrast), and its control holds the model fixed, ours the bit width. Replacing one action at a restored state and letting the original policy continue (our "this step only" contrast) locates decisive errors (Zhang et al., 2025), attributes failures (Shah, 2026) and yields training signal (Li et al., 2026a,b); the factorial design itself is standard.

**Precision routing, escalation and their signals.** DMR (Li et al., 2026c) learns a per-step BF16/W4 router from KL supervision and episode reward; sensitivity has been localised to phases (Lu et al., 2026), task types (Zhang et al., 2026) and high-entropy, high-KL tokens (Lotfi et al., 2026). Agents also defer single decisions when uncertain (Piatrashyn et al., 2026), learn cost-aware escalation (Lei and Huang, 2026), or are routed per step against execution-verified labels (Yang et al., 2026). Continuing another model's trajectory has its own cost (Ganz et al., 2026); a policy's confidence does not reliably locate consequential steps under executed replay (Zhang, 2026); and KL measures how much a quantized model disagrees, not in which direction (Nikolić et al., 2026). Our evaluation measures action-only, continuation-only and joint substitutions on shared states, using rescue–damage accounting and reference interventions to interpret their task value.

**Evaluation noise and user simulators.** Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

## 3 Measuring the Task Value of an Upgrade

### 3.1 Setting

**Tasks and agents.** We use $\tau^2$-bench in text mode: a development panel of 58 retail and 27 airline tasks plus 80 telecom development tasks, and a holdout of 34 telecom tasks that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8); a second family, Mistral-Small-3.2-24B, is a scope check (Table 1). One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success is

the final-database match with normal termination, without all of $\tau^2$-bench's official communication checks; telecom uses the official task-basis success. BFCL multi-turn and ALFWorld run once per item without a user simulator (Appendix A).

**Quantization and serving.** Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main configurations use GPTQ (Frantar et al., 2023) and AWQ-style scaling (Lin et al., 2024) at W4 (group size 128), with a BF16 KV cache, calibrated on BF16 agent trajectories of 57 retail and airline tasks disjoint from the development tasks. Separate KV8 configurations quantize the cache to INT8 while retaining BF16 weights. Stress configurations (GPTQ-W3, RTN-W3, rotated KV4) and QuaRot, SpinQuant and KIVI (Ashkboos et al., 2024; Liu et al., 2025, 2024) are scope checks (Table 1; Appendix Table 4). B and Q run on vLLM (Kwon et al., 2023) with batch-invariant kernels (He and Thinking Machines Lab, 2025) and deterministic compilation in separate server processes (no shared prefix cache). A configuration pairs a model with a quantization setting. A *block* is one closed-loop run of Q over the task pool under one prompt nonce (a random session-identifier line in the system prompt); blocks are the replicates, and a *unit* is one task in one block.

### 3.2 Four branches on one state

**Frozen state.** Before Q's decision $t$ we restore the state $s_t$: the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q act from identical prompt token ids, and the state is written to disk before any branch runs.

**Four branches.** $Y_{\mathrm{XY}} \in \{0, 1\}$ is task success when the action at $s_t$ comes from X and model Y executes the rest of the episode, X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$; all four branches share the remaining budget, and each upgrade option is a paired contrast with QQ: $G$ (this step only), $H$ (from the next step) and $F$ (both), defined in Figure 1. A *rescue* is a unit where the upgraded branch succeeds and QQ fails, a *damage* the reverse; *turnover* is rescues plus damages, *net* rescues minus damages. An outcome is *continuation dependent* when the action swap's effect differs between Q's and B's continuation. All contrasts measure effects on task success, not which action is correct (symbols: Appendix A.1).

**First divergent state**
First decision where B and Q differ in action kind, tool calls or normalised arguments. Wording alone does not qualify.

**Ordinary sampled state**
One decision per task–block unit, selected by a fixed hash before generation. Sampling does not depend on disagreement or outcome.

**Common state**
An ordinary sampled state where B, Q and Q′ all act on exactly the same history.

Figure 2: State selection defines the question. First divergences measure the consequences of a disagreement; ordinary states measure interventions without selecting disagreements. Common states compare action sources directly.

**The action being replaced.** The intervention replaces a complete assistant turn, including its language and structured tool calls. The chosen message and tool results enter the subsequent history; the continuing model makes every later decision. The contrast therefore measures the task consequence of changing execution at that state, rather than an intrinsic correctness label for the action.

**Reference interventions.** A large turnover could mean that a decision needs higher precision, or only that it is sensitive to a change of model. We therefore rerun the design with Q′ in place of B: a second GPTQ-W4 checkpoint of the same model, calibrated on a re-drawn state and row sample of the same calibration trajectories, with the same Q episodes, tasks and nonce blocks and a protocol frozen before any outcome (q4 and q8 GPTQ-W4; GPTQ-W3 as a stress check). §5 adds common states and a key-order change of Q's own action.

### 3.3 Making the counterfactuals auditable

**Replay acceptance.** Before any development outcome was analysed, exact no-op replays on real divergence states passed a gate, and in production every fork task re-ran the QQ no-op on an outcome-independent hash sample of its states; all sampled states of every state source and of the same-precision run passed (Appendix A). Gates passed 20/20 Q-to-Q and 10/10 B-to-B replays; production checks passed 924/924 develop-

ment states. Exact replay is a tested property of this stack (greedy decoding, vLLM 0.29.0, batch-invariant kernels, deterministic compilation, dense BF16 kernels), not a general one (Yuan et al., 2025); determinism tests and untested conditions are in Appendix A.

### 3.4 Which states

**State sources.** A *common state* is an ordinary sampled state at which several action sources are compared; Figure 2 distinguishes the state sources. Full-pool rates include every task–block unit, counting no divergence as zero effect; mode and pattern analyses condition on divergent states. Contrasts are paired within a configuration and descriptive across configurations, which select their own divergences; only complete blocks enter.

**Intervals and interpretation.** Fixed-pool intervals describe variation across nonce blocks on the evaluated tasks; crossed task–block intervals also resample tasks and address a new-task target. We call an effect unresolved when the interval used for its reading includes zero; this does not establish equivalence. Figure captions identify the reading interval, and the appendix gives the full inference rules. Each branch measures one realised outcome under the fixed greedy execution stack, not an expectation over user simulators or perturbations.

## 4 Result 1: An Upgrade Has Two-Sided Consequences

The holdout is pre-registered; the upgrade-mode and pattern analyses are exploratory and post hoc.

**Large outcome changes, small net estimate.** On the $\tau^2$-bench development panel (85 tasks $\times$ 20 complete blocks), replacing a W4 agent's action at the first divergent state with B's, while Q continues, changes 15.5–20.9% of task outcomes, with rescues and damages of similar size (Figure 3a). Two fixed-pool intervals of the net exclude zero (negative for q4 GPTQ-W4, positive for q8 GPTQ-W4); no new-task interval does. Pooled over the four W4 configurations, the net is $+0.7$ pp (fixed pool $[-0.3, +1.6]$; new tasks $[-2.1, +3.4]$). The net is a local replacement effect, not an end-to-end comparison (Appendix A.2).

**A negative holdout estimate.** On 34 unseen telecom tasks, under a protocol frozen before any holdout unit ran (12 fresh blocks, three W4 configurations), the replacement again rescued and damaged units (Figure 3b). The primary net, $-4.57$ pp (crossed 90% $[-7.17, -1.96]$), lies below zero. The frozen readings had only positive-gain and small-effect categories, a design omission, so the protocol classifies the result as *inconclusive* (Figure 3b; Appendix C). Both pre-registered secondary checks passed: S1 (rescues and damages both positive) and S2 (cross-half repeatability of task fragility, $\rho = 0.588$, one-sided permutation $p = 0.0001$). A post hoc QQ no-op replay of all 1202 divergent states, under a protocol frozen before any replay ran, reproduced every original episode, so the baseline shared by all holdout contrasts is not a replay artifact. The holdout changes domain as well as tasks (Appendix C.1).

**The intervention defines which rescues count.** A joint rescue succeeds only when both the current action and the continuation change (Figure 1); action-only evaluation misses it. Conversely, a rescue under Q's continuation can disappear when B also takes over, so whole-takeover evaluation can miss a useful action-only replacement. These observed patterns illustrate why $G$, $H$ and $F$ are separate targets. Their full counts and permutation references are exploratory and appear in Appendix B.2; a gain measured for one mode cannot be reused as the gain of another.

## 5 Result 2: Flips Alone Do Not Diagnose a Need for More Bits

A flip shows that an outcome is sensitive to substitution. To ask whether it indicates a need for higher precision, we compare BF16 with two reference interventions (Figure 4). Their populations and comparison targets differ, so each reference answers a distinct question.

**A same-precision checkpoint also changes outcomes.** Q′ replaces the current action without increasing bit width. On the matched W4 task–block panel it produces 17.7% turnover, compared with 16.8% for BF16, with rescues and damages in both directions (Figure 4a). Changing the W3 checkpoint produces the same qualitative pattern. These are observed turnover estimates; equivalence within $\pm 3$ pp is not established.

**An identical tool operation can lead to a different outcome.** At 300 eligible q8 states, we reverse only the order of Q's tool-argument keys. Every perturbed call parses to the same arguments and produces the same environment effects;

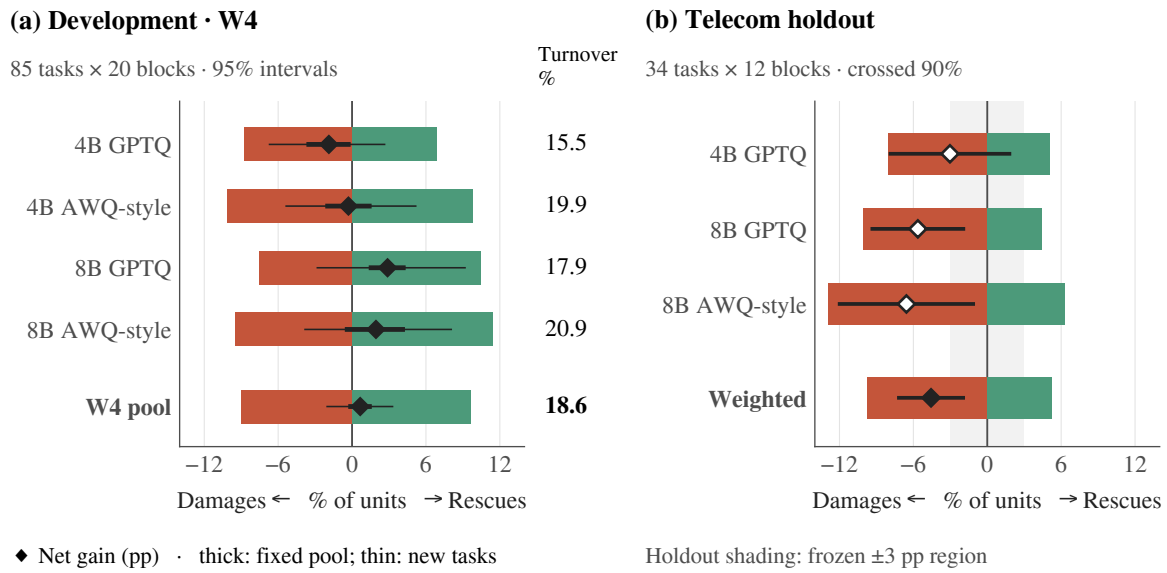



Figure 3: Action-only substitution at the first tool-structure divergence. Bars show rescues (green) and damages (red); diamonds show net gain, all per full-pool task–block unit. (a) Development: thick fixed-pool and thin new-task 95% intervals, with turnover at right. (b) Holdout: crossed 90% reading intervals; shading marks the frozen ±3 pp region. Hollow configuration diamonds are descriptive; the weighted estimate is primary. AWQ means AWQ-style scaling. Detailed mode and pattern comparisons appear in Figure 6.

the agent subsequently reads a different token sequence. Outcomes change on 6.7% of these states, versus 9.0% when replacing the action with BF16's (Figure 4b). The paired turnover difference is −2.3 pp [−8.4, +2.9], leaving the difference unresolved. This exploratory reference covers one representation change on 41 tasks, rather than a general perturbation distribution (Appendix D.3).

**Shared states isolate the additional value of BF16.** The preceding design selects each substitute's own first divergence. It compares two selection-and-replacement protocols, rather than two actions at the same decision. We therefore compare B and Q′ directly at 510 shared ordinary q8 states. BF16's extra action value is +1.0 pp under Q's continuation (two-look 97% block-$t$ interval [−4.2, +6.2]) and +0.8 pp under B's ([−2.2, +3.8]; Figure 4c). Both comparisons remain unresolved and neither establishes equivalence. The six-block extension and two-look rule were fixed after the unresolved first look; the timeline, bootstrap intervals and sensitivity excluding earlier studied states are in Appendix D.2.

Together, the references establish a limit on interpreting flips: a changed outcome does not identify a decision that needs higher precision. The additional task value of BF16 requires a direct paired comparison of actions under the continuation that will run.

## 6 Result 3: Finding Changes Is Not Selecting Gains

**Selecting outcome-changing decisions.** $G$, $H$ and $F$ require running B; a deployable rule sees only Q. On one development block, Q's token uncertainty separates states where B and Q act differently from states where they agree, but no Q-visible statistic reliably separates flips from non-flips among the former (Appendix E.1). We then froze a Q-only trigger on that block, before any later state was scored: escalate one decision (B replaces only the current action and Q continues: the $G$ contrast) when the minimum Q margin over its action is low (margins predict cross-precision divergence; Du et al., 2026), with per-configuration thresholds targeting 5, 10 and 20% on the development block (Figure 5). These frozen thresholds give selection rates rather than exact validation quotas; random and oracle baselines match the actual counts. The margin score was chosen among three Q-only candidates (entropy, margin, mean log-probability) by its net gain over random on that block. On two later blocks of the development tasks (not the telecom holdout), at ordinary states from Q's own trajectories (six configurations: four

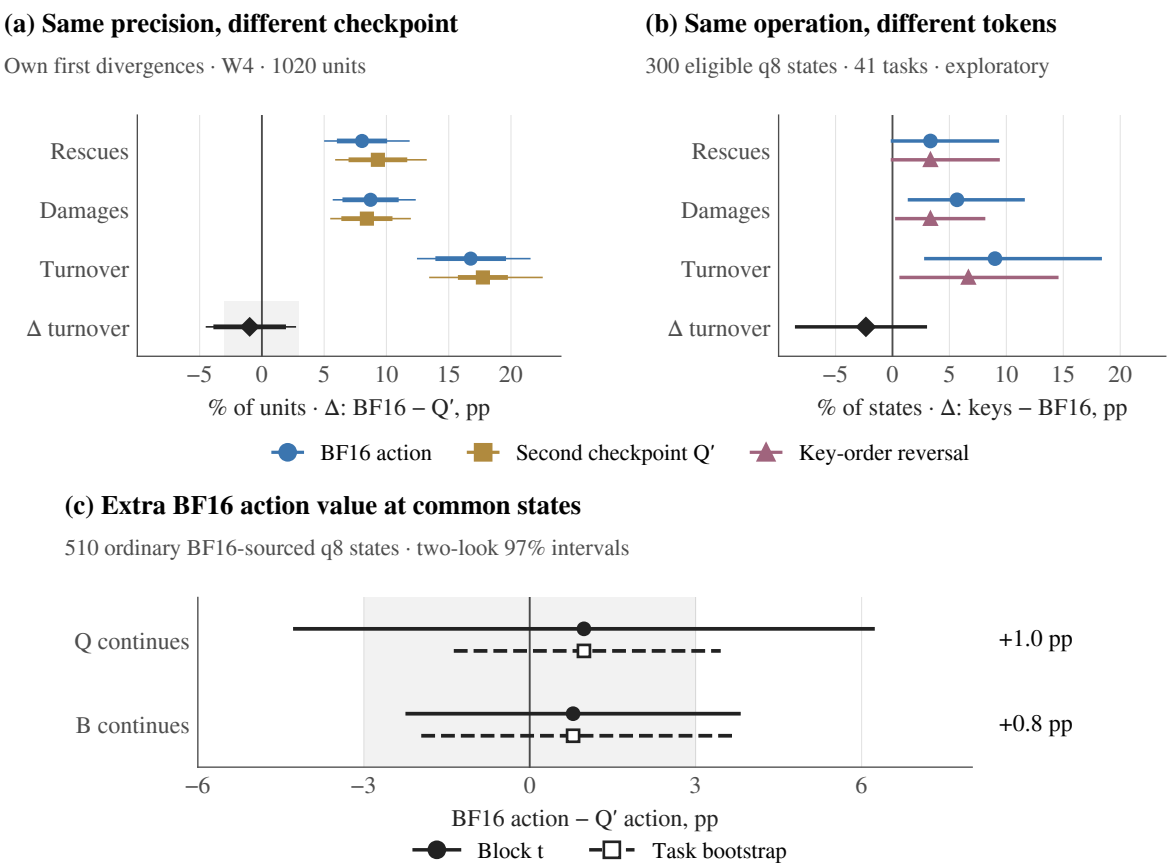



Figure 4: Three references for interpreting an upgrade. (a) Same-precision checkpoint substitution at each design's own first divergences: thick fixed-pool and thin new-task 95% intervals. (b) Key-order reversal on eligible tool calls: the wider of block and crossed 95% intervals. (c) Extra BF16 action value at shared states: two-look 97% block-$t$ and task-bootstrap intervals. Shading marks $\pm 3$ pp for paired differences. Populations are separate; reference turnover is not subtracted from BF16 turnover. Full values, other intervals and protocol timelines: Appendix D.1 and Table 13.

W4, two KV8; 1020 states, 69 outcome-changing) its capture of outcome-changing states exceeded matched-count random selection at 5 and 20% (exact one-sided $p = .0182$ and .0002; $p = .0613$ at 10%; Figure 5a). At ordinary states from BF16 trajectories (secondary), only the 20% rate enriches (Appendix E.1).

**Testing benefit against random selection.** The captured states were rescues and damages alike (Figure 5a). Against matched-count random selection, its net gain has exact one-sided $p = .0345$, .4169 and .2617 (Figure 5b). No escalation rate was designated primary in advance, and the 5% result does not survive Holm correction over the three rates. A hindsight oracle with the same per-configuration counts nets far more (Figure 5b), positive by construction because it picks only rescues. A post hoc $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ score, which needs BF16, also captures turnover above chance on these states without a significant net gain (exploratory, post hoc; Appendices E.2 and E.3).

**What this test establishes.** Enriching outcome-changing decisions is not enough for reliable net gain (cf. Zhang, 2026). The later blocks reuse the development tasks, and only this simple frozen rule is tested. The result does not test learned or other uncertainty-based escalation (Li et al., 2026c; Piatrashyn et al., 2026; Lei and Huang, 2026) and does not refute sensitivity routing in general.

## 7 Discussion

**Sensitivity and value answer different questions.** The two reference interventions are central to interpreting an upgrade. Q′ shows that higher precision is not necessary for two-sided outcome changes. Key-order reversal shows that some changes arise even when the executed tool

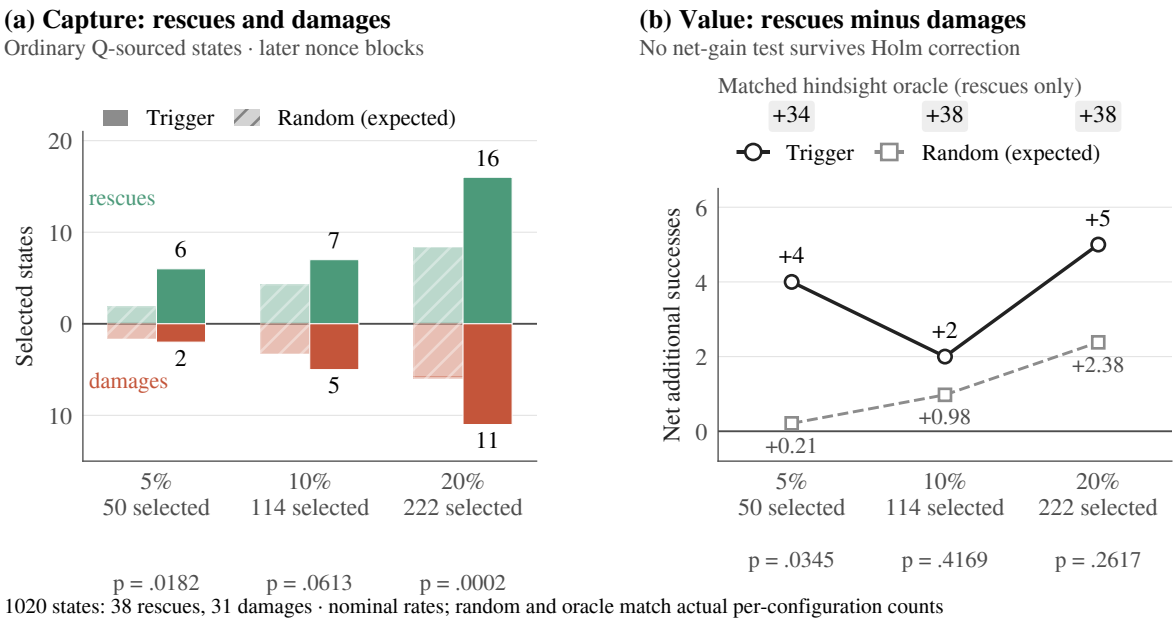



Figure 5: One frozen Q-only margin trigger on 1020 ordinary Q-sourced validation states. (a) Selected rescues and damages versus matched-count random selection. (b) Net additional successes; the grey strip gives the matched-count hindsight oracle, which selects rescues by construction. Nominal rates are shown with actual selected counts. Exact one-sided $p$-values compare capture in (a) and net gain in (b); no net test survives Holm correction across the three rates. Full values: Table 14.

operation stays identical. Neither reference establishes that precision is irrelevant, and neither is a noise floor to subtract from BF16's turnover. The common-state experiment asks the direct value

| Scope and population | Task consequences | Interpretation |
|---|---|---|
| **BFCL multi-turn** | | |
| q4/q8 W4; no simulated user | Turnover 5.2–15.2%; pool net $+0.1$ $[-1.0, +1.2]$. The q4 AWQ-style net is $-3.8$ $[-6.5, -1.3]$. | Two-sided changes in a tool benchmark. |
| **ALFWorld** | | |
| q8 W4; 134 games; no simulated user | Turnover 21.6–24.6%; pool net $-3.7$ $[-9.7, +1.9]$. | Two-sided changes in an interactive world. |
| **Mistral-Small-3.2-24B** | | |
| GPTQ/RTN-W4; 83 tasks; ordinary BF16-sourced states | GPTQ rescues 4.6 and damages 5.4 pp; every mode's net remains unresolved. | A second family; a different state population. |
| **Lower-bit stress** | | |
| q8 RTN-W3; 255 first divergences; 3 blocks | Action-only net $G_{\text{all}} = -0.4$ $[-3.8, +3.0]$, new-task $[-4.7, +3.9]$. On divergent states, $H = +15.3$ and $F = +18.4$ [9.4, 28.2] pp. | Takeover can help. Selected from five stress configurations; exploratory. |
| **QuaRot / SpinQuant** | | |
| q8 W4A4; ordinary BF16-sourced states; 3 blocks | KL 0.165 vs 0.199; task-value contrast $+1.2$ $[-4.7, +7.1]$. | Lower KL without a resolved value ordering. |
| **KV-cache precision** | | |
| KV8 first divergences; KIVI ordinary states | KV8 turnover 6.5–8.1%; KIVI-2/4 versus KV8 value contrasts remain unresolved. | Scope beyond weight-only quantization. |

Table 1: Scope checks on separate populations (Appendix F). BFCL and ALFWorld use post hoc item-bootstrap 95% intervals. QuaRot/SpinQuant and RTN-W3 $F$ use crossed 95%; RTN-W3 action-only net gives fixed-pool and new-task 95%. None of these panels supplies a Q′ reference: the separate W3 reference is GPTQ-W3 (Appendix D.1), rather than RTN-W3.

question, but its intervals still permit a useful BF16 advantage. An unresolved comparison therefore warrants neither automatic escalation nor a claim that escalation cannot help.

**Where takeover can help.** The exploratory RTN-W3 example in Table 1 illustrates the importance of specifying the intervention: action-only replacement has a near-zero estimate, whereas continuation takeover and joint replacement have positive gains. This selected, exploratory stress example illustrates a benefit for a specific continuation intervention; the selection and population are stated in the table. It nevertheless shows why a null action-only result cannot settle the value of a longer takeover. The scope checks cover additional models, environments and quantization settings, each on its own state population.

**What a selector needs to predict.** A sensitivity score can find disagreements or outcome changes while selecting rescues and damages together. Our frozen margin rule demonstrates this distinction on later nonce blocks of the same tasks. Selection labels should match the intervention a deployment will execute. An action-only rule and a continuation router therefore need different outcome targets. Learned selection and cost-aware routing are natural extensions: evaluation would compare signed task benefit at a shared BF16 compute budget, adding deployment costs to the task-value measurements made here.

**Conditional task value.** Each rescue or damage is a paired, realised consequence under the specified greedy stack. Exact replay establishes the consistency of that comparison. The key-order result illustrates its dependence on the representation the agent reads; changing the user or prompt defines another execution condition. These labels describe consequences in the measured setting.

**Value across execution conditions.** To evaluate an upgrade over a deployment distribution, specify the prompt and user conditions and repeat paired branches across them. Averaging the resulting success differences defines an expected task value. Label stability and average benefit then become separate questions; the present fixed-stack measurements provide the starting point for that evaluation.

**Recommendations for evaluation.** Report the rescues, damages and net gain of the actual upgrade mode on paired states. Include reference substitutions to test whether sensitivity alone supports the intended precision claim. Distinguish the fixed task pool from a new-task target, and evaluate a selection rule by net benefit at matched counts. Table 1 summarises the present coverage; detailed inference and protocol amendments are in the appendix.

## 8 Conclusion

Precision upgrades have an intervention-specific task value. Our audited forks separate replacing an action from taking over the continuation, while reference substitutions show why outcome changes alone cannot establish a need for more bits. On the main W4 panel, rescues and damages largely cancel; the common-state comparison leaves BF16's additional value unresolved. One frozen uncertainty rule enriches outcome changes without reliable net benefit. Together, these results motivate evaluating fallback decisions through paired task consequences, reference interventions and the net gain of the policy actually executed.

## Limitations

**Inference scope.** The contrasts measure realised task effects under the fixed execution stack. Fixed-pool and new-task intervals address different targets; intervals containing zero leave an effect unresolved, while equivalence requires the designated equivalence test. $G$, $H$ and $F$ are offline intervention values rather than the gain of a deployed selection policy. The outcome-class and label-permutation analyses are exploratory and post hoc (Appendix B.2).

**Two model families, limited coverage.** Most crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. The second family, Mistral-Small-3.2-24B (GPTQ-W4, RTN-W4), was forked only at ordinary BF16-sourced $\tau^2$-bench states of an 83-task conditional population (Appendix F.3); it has no first-divergent-state forks, no same-precision reference and no outcome-class analysis. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix A.3). A time-boxed Llama-3.1-8B-Instruct check (GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no evidence for that family.

**Simulated quantization.** Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

**One user simulator; two small controls.** All $\tau^2$-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The first-divergent-state control has 20 states per configuration, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw: the replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the ordinary-state control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation; the first-divergent-state control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel (six blocks) and GPTQ-W3 on three blocks, and swaps one calibration sample for another, re-drawn from the same calibration trajectories; it does not cover checkpoints calibrated on independent datasets. Six is the longest complete common prefix under the frozen rule, because the seventh block of the same-precision q8 design is incomplete. It is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8, RTN or rotated KV4.

**Small confirmation and partial state sets.** The pre-registered holdout covers 34 telecom tasks and three configurations. Its pre-registered readings had only positive-gain and small-effect categories; omitting a negative-effect category was a design omission, and it leaves the protocol able to classify the negative estimate only as inconclusive. It applies only to telecom, these three configurations and this simulator. A post hoc replay of all 1202 divergent states reproduced every original QQ episode (Appendix C); it rules out a replay artifact, not other causes of the negative estimate. Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout configurations show no detectable net effect (Appendix C.1), but their 6-block intervals are wide and q4 has no telecom development forks. These data cannot separate a telecom domain effect from heterogeneity among the 34 holdout tasks. First-divergent-state forks intervene only at the first tool-structure divergence per task and block. Ordinary-state forks sample one state per task and block; their main analyses use three blocks, so their block intervals rest on two degrees of freedom, and the later panels use two to six blocks. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

## Ethics Statement

This work involves no crowdworkers, no human annotation and no personal data. Where a user is present ($\tau^2$-bench), it is simulated by an LLM; tasks are public benchmark tasks, and all agents are openly released Qwen3, Mistral-Small-3.2 and (gate check only) Llama-3.1 checkpoints, or simulated-quantized versions of them. Outcomes use the automatic task-success endpoints defined in §3.1, including the database-only retail and airline checks, and action categories are assigned by fixed rules; reported values map to committed analysis outputs (Appendix G). The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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## A Setup, Replay Acceptance and Notation (supports §3)

This appendix supports §3: it gives the splits, how a state is restored and forked, which replay checks

passed on this serving stack, and the internal name behind each main-text term.

**Splits.** The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from $\tau^2$-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

**Success endpoints.** For retail and airline we use the final-database match with normal termination. This is not $\tau^2$-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures, except for the explicitly amended missing-data rules of the Mistral panel (Appendix F.3). Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

**Fork mechanics.** The forced action is returned by a fork agent for the one step that follows $h_t$. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. $\tau^2$-bench resets step and error counters when it starts from a history. The fork restores them to the values used at $t$, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

**Replay acceptance details.** In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Besides the 924 sampled development states, 328/328 sampled Bank U states passed the QQ no-op, and the Bank U gate passed a further 135/135 no-ops (a different set from the holdout's 135 QQ no-ops, all of which also passed; a later full replay of all 1202 holdout divergent states also matched; Appendix C). Serving in separate processes closes the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.

**Determinism of the serving stack.** Closed-loop determinism pairs, the same configuration on two nodes at different times, reproduced whole episodes token for token for dense BF16 paths at tensor parallelism 1 and 2 with 1 to 16 concurrent requests (e.g. 257/257 and 1520/1520 episodes). Nine arms that two studies ran independently at the same states were byte-identical. We did not test exactness under KV-cache preemption, or on other models, GPUs or vLLM versions. A deployed AWQ kernel path with fp16 dequantization was not replay-exact, and no reported outcome uses it.

**User simulator.** The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

### A.1 Internal names and notation

**Glossary.** The main text uses plain terms; this supplement also uses the internal names of the analysis code. A *first divergent state* is a *Bank P* state. An *ordinary sampled state* is a *Bank U* state; its sub-banks are drawn from Q's own trajectories (U1) or from trajectories shared by all configurations of a model (U2B: BF16 trajectories; U2Q: Q trajectories). A *common state* is an ordinary state of the common-state design (pre-registered at three blocks, extended to six by amendment), at which B, Q and Q′ each supply the current action. A *cell* is a configuration, named by model (q4: Qwen3-4B; q8: Qwen3-8B; q14: Qwen3-14B; m24: Mistral-Small-3.2-24B) and quantization setting. Nonce blocks carry ids such as fNN (development) and qN (stress and W3 panels); the holdout blocks have their own ids. Branches are written by action source and continuing model (BB, BQ, QB, QQ; KQ is the key-order branch, Appendix D.3). Symbols: $r$ and $d$ are the rescue and damage rates of the action swap, $M = r + d$ is the turnover and $G_{\text{all}} = r - d$ the net gain on the full pool; $\Delta_{\text{Q}}$, $\Delta_{\text{B}}$ and the off-diagonal share are defined below; $I = \Delta_{\text{Q}} - \Delta_{\text{B}}$ is the interaction of action and continuation (Table 6); $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ compares methods on shared ordinary states (Appendix F.5); $K(c)$ is B's action value beyond Q′'s at a common state under continuation $c$ (Appendix D.2); $n_{\text{blk}}$ is the number of complete blocks.

**Action swap, rescues and damages.** The paired effect of the action swap under each continuation

is $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ ($= G$) and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$, each in $\{-1, 0, +1\}$. When $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$ ("off-diagonal"), the swap's value depends on the continuation. For the action swap, a *rescue* is $(Y_{\mathrm{BQ}}, Y_{\mathrm{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates $r$ and $d$, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

**Same-precision notation.** $M'$, $r'$, $d'$ and the off-diagonal share are defined likewise with Q′ for B.

**Which states.** *Bank P*: for each task, block and cell, the first decision at which $a_{\mathrm{B}}$ and $a_{\mathrm{Q}}$ differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks (up to six for U2B in §5–§7), from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

**Denominators and inference.** We keep three denominators separate. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; $r$, $d$, $M$ and $G_{\mathrm{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; $G$, $H$, $F$ and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. With $n_{\mathrm{blk}}$ complete blocks (only complete blocks enter), the primary *fixed-pool* interval treats the task pool as fixed (block bootstrap if $n_{\mathrm{blk}} \geq 10$, else block-$t$, a $t$ interval over block means); the *new-task* interval is a crossed task×block bootstrap ("crossed"). We call a contrast *resolved* under an interval when that interval excludes zero and name the interval; "not resolved" means the reported interval or intervals include zero. Where a frozen protocol sets its own reading rule, we state it with the result.

### A.2 Separating local value from end-to-end success

The replacement gains $G_{\mathrm{all}}$, $G$, $H$ and $F$ are not end-to-end comparisons. Table 2 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

### A.3 Coverage Matrix

Table 3 lists the completed crossed-intervention coverage behind every Qwen3 Bank P and Bank U result; the coverage of the U2B panels (§6, §7) is in Appendix F.5. It counts tasks, states and complete blocks after the strict admission rule. Four of the five stress cells turn over 18.4–20.4% of units with three blocks; we keep them separate from W4, and no W2 crossed-intervention panel is reported.

### A.4 Quantization configurations

Table 4 gives the quantization settings of every checkpoint configuration, read from the checkpoint metadata and the quantization code. Not recorded: the AWQ-style row-sample size (4096 is the code default) and the KV-cache scale dtype (the hook computes in fp32); Mistral has no second calibration draw.

## B Two-Sided Consequences in Full (supports §4)

This appendix supports §4: replacing a W4 agent's action produces both rescues and damages, and a gain measured for one upgrade mode cannot be reused as the gain of another.

Table 5 gives every cell of Figure 3a, including the stress cells, with new-task intervals. Table 6 gives $G$, $H$, $F$ and their paired differences for every cell, with $F - G$, $F - H$ and the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ summaries.

### B.1 New-task intervals for the four-branch estimands

Table 7 gives all new-task intervals corresponding to Table 6.

### B.2 Outcome Classes and a Label-Permutation Reference

Everything in this section is exploratory and post hoc. It uses the four outcomes of each divergent state in the order BB, BQ, QB, QQ; $k$ is the number of successful arms.

**Five classes.** The 16 patterns fall into five classes: unchanged (`0000`, `1111`); action only (`1100`, `0011`: the outcome follows the action under both continuations); continuation only (`1010`, `0101`); one arm differs from the other three ($k$ odd); strict reversal (`1001`, `0110`). Changing the action changes the outcome under at least one continuation exactly in the action-only, one-arm and

| Cell | Q / BF16 (%) | Q−BF16 [95% CI] | NI 3 pp | TOST ±3 pp |
|---|---|---|---|---|
| q4 GPTQ-W4 | 40.0 / 35.8 | +4.2 [−2.4, +10.5] | pass | fail |
| q4 AWQ-style W4 | 36.8 / 35.8 | +0.9 [−5.8, +7.5] | fail | fail |
| q8 GPTQ-W4 | 30.1 / 30.4 | −0.2 [−7.7, +6.8] | fail | fail |
| q8 AWQ-style W4 | 31.5 / 30.4 | +1.2 [−5.3, +7.6] | fail | fail |
| pool, all four | | +1.5 [−2.7, +5.6] | pass | fail |

Table 2: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

| Cell | $\tau^2$-bench retail (Bank P) | $\tau^2$-bench airline (Bank P) | telecom dev (Bank P) | Bank U (retail / airline) | BFCL / ALFWorld |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | 58 t · 727 st · 20/20 b | 27 t · 375 st · 20/20 b | holdout only | 3 b · 393 / 177 rows | 788 items · 573 st / – |
| q4 AWQ-style W4 | 58 t · 986 st · 20/20 b | 27 t · 453 st · 20/20 b | – | 3 b · 393 / 177 rows | 788 items · 642 st / – |
| q4 KV8 | 58 t · 412 st · 22/22 b | 27 t · 181 st · 22/22 b | – | 3 b · 393 / 177 rows | – / – |
| q8 GPTQ-W4 | 58 t · 1017 st · 20/20 b | 27 t · 456 st · 20/20 b | 80 t · 477 st · 6/6 b; holdout | 3 b · 393 / 177 rows | 788 · 533 st / 134 · 115 st |
| q8 AWQ-style W4 | 58 t · 1087 st · 20/20 b | 27 t · 475 st · 20/20 b | 80 t · 480 st · 6/6 b; holdout | 3 b · 393 / 177 rows | 788 · 548 st / 134 · 119 st |
| q8 KV8 | 58 t · 585 st · 22/22 b | 27 t · 270 st · 22/22 b | – | 3 b · 393 / 177 rows | – / – |
| q4 GPTQ-W3 | 58 t · 162 st · 3/3 b | 27 t · 70 st · 3/3 b | – | – | 591 · 540 st† / – |
| q4 KV4-rot | 58 t · 164 st · 3/3 b | 27 t · 69 st · 3/3 b | – | – | – / – |
| q8 GPTQ-W3 | 58 t · 166 st · 3/3 b | 27 t · 76 st · 3/3 b | – | – | 788 · 696 st / 134 · 130 st |
| q8 KV4-rot | 58 t · 165 st · 3/3 b | 27 t · 74 st · 3/3 b | – | – | – / – |
| q8 RTN-W3 | 58 t · 174 st · 3/3 b | 27 t · 81 st · 3/3 b | – | – | – / – |
| q4 / q8 KV4 (unrotated) | – | – | – | – | 788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st |
| q14 GPTQ-W4 | – | – | – | – | 788 · 471 st / – |

Table 3: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix F.2). "holdout": the cell is in the telecom holdout (Appendix C). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix C.1) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix F.4). The end-to-end q4 AWQ-W3 run is incomplete (two airline episodes hit 600 s serving timeouts), and so are the q4 RTN-W2 stress runs (context-window overflows of the degenerate model and a truncated result file); neither is reported.

reversal classes, and changing the continuation exactly in the continuation-only, one-arm and reversal classes. Both change it exactly on the off-diagonal states, so the two shares overlap and must not be added, and their gap equals action only minus continuation only: it comes from $k = 2$ states alone and restates the $k = 2$ arrangement below on all divergent states. Two named patterns follow the upgrade modes: joint rescue (1000: only the joint upgrade succeeds) and rescue undone ($Y_{\mathrm{BQ}} = 1$, $Y_{\mathrm{QQ}} = 0$, $Y_{\mathrm{BB}} = 0$: B's action rescues under Q's continuation and B's takeover of the continuation undoes it). Table 8 gives the shares. The action-side gap holds in the $\tau^2$-bench retail/airline W4 and KV8 pools, in the W4 same-precision fork and in the BFCL W4 pool. It is near zero on the telecom development tasks (+0.1 [−2.1, +2.4] pp, 6-block $t$) and in the q4 GPTQ-W3 and KV4-rot stress cells, and it is not resolved in the ALFWorld W4 pool (+3.4 [−1.8, +8.4]) or

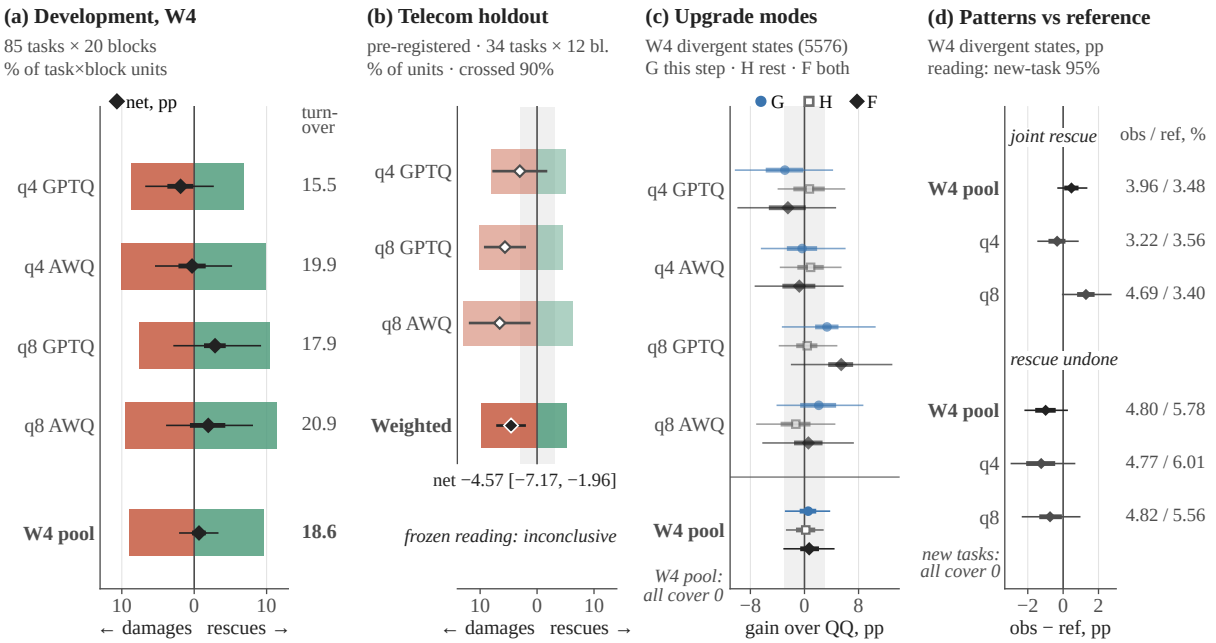



Figure 6: Complete consequence display: (a,b) full-pool development and holdout estimates; (c) action-only, continuation-only and joint effects on divergent states; (d) joint rescue and rescue undone relative to label permutation. Panels (c,d) are exploratory and post hoc. Thick: fixed-pool 95%; thin: new-task 95%; holdout uses crossed 90%. Each population remains separate. Values: Tables 5, 11, 6 and 9.

the W3 same-precision pair (+1.9 [−2.9, +6.7]). Its point estimate is negative in the q8 RTN-W3 stress cell (−7.8; 3-block $t$ [−18.1, +2.4], new-task [−15.3, −1.2]) and, not shown, in ALFWorld q8 GPTQ-W3 (−19.2 [−27.3, −10.8]). In the ALFWorld and BFCL KV4 cells (not shown) Q almost never succeeds, so continuation-only states dominate there mechanically.

**W4 pool details.** In the $\tau^2$-bench W4 pool, the gap between the action and continuation shares is +5.2 pp (new-task [+3.1, +7.7]): action effects are somewhat more common, and the continuation changes the outcome mostly where the action matters too. Of the 26.9% off-diagonal states, 23.8% have one arm differing from the other three and 3.2% are strict reversals; states where the swap helps more under Q ($I > 0$, 13.6%) and under B ($I < 0$, 13.4%) are about equally frequent, so the mean interaction is +0.0 [−1.9, +2.0] pp. With fixed-pool, then new-task 95% intervals, changing the action changes the outcome under at least one continuation in 34.9% ([33.6, 36.0]; [28.8, 41.0]) of W4 divergent states, and changing the continuation controller does so under at least one action in 29.6% ([28.5, 30.7]; [24.2, 35.3]). The two sets overlap in the 26.9% ([25.7, 28.1]; [22.1, 32.0]) where the swap's effect depends on the continuation, so the shares must not be added; all four arms agree in 62.4%.

**A label-permutation reference.** Given $k$, assigning the four outcomes to the four arms uniformly at random gives the off-diagonal share $E_0 = P(k \text{ odd}) + P(k = 2)/3$. In every cell the observed share minus $E_0$ equals $P(k = 2)\,[P(\text{reversal} \mid k = 2) - 1/3]$, so it can depart from $E_0$ only through strict reversals. In the W4 pool the share is 26.9% against $E_0 = 28.4\%$ (−1.4 [−1.8, −1.0] pp, fixed pool, block bootstrap over 20 blocks; KV8: 15.3 against 18.9%), because reversals are 23.6% [19.2, 28.3] (task bootstrap) of $k = 2$ states instead of one third. This shows only that strict reversals are rarer than under random labelling. It does not show that the continuation controller is irrelevant, which would require $Y_{\text{BB}} = Y_{\text{BQ}}$ and $Y_{\text{QB}} = Y_{\text{QQ}}$, or that the interaction is zero: the reference itself allows $I \neq 0$ in most arrangements. Conversely, an off-diagonal state does not show a distinct interaction mechanism: if an arm succeeds when an action score plus a continuation score exceeds a threshold, exactly additive scores can make only BB succeed (`1000`), although they never produce a strict reversal.

**Joint rescue and rescue undone against the reference.** The same reference, applied to the two named patterns, keeps $k$ for every state and permutes the four outcome labels uniformly: `1000` then has probability $1/4$ at $k = 1$, and rescue undone (`0100` or `0110`) has $1/4$ at $k = 1$ plus $1/6$ at $k = 2$, so the reference shares are $P(k{=}1)/4$ and $P(k{=}1)/4 + P(k{=}2)/6$, computed on the same states and bootstrap draws as the observed shares (Table 9). In the $\tau^2$-bench W4 pool, joint rescue exceeds the reference by $+0.47$ pp on the fixed task pool ([$+0.06$, $+0.87$]), but the new-task interval spans zero ([$-0.33$, $+1.37$]); rescue undone is less frequent than the reference ($-0.99$ [$-1.58$, $-0.42$]; new tasks [$-2.19$, $+0.27$]). The excess of joint rescue is concentrated in the q8 cells, and no group shows a new-task excess of rescue undone. Because the reference keeps $k$ and does not model execution, a share near it says that the arrangement is about as common as a random arrangement of the same number of successes, not that the outcomes are noise or that the continuation is irrelevant. The observed shares in the W4 pool, with both intervals, are 4.0% for joint rescue (fixed pool [3.5, 4.5]; new tasks [2.7, 5.3]) and 4.8% for rescue undone ([4.1, 5.4]; [3.2, 6.5]).

**The $k = 2$ arrangement.** At $k = 2$ the two successes share the action in 56.4% of W4 states and 80.3% of KV8 states, against one third under random labelling. A statement that outcomes are mainly determined by the action holds only within this subset, which is 13.8% (W4) and 14.5% (KV8) of divergent states.

### B.3 Task fragility

All analyses in this subsection are exploratory and post hoc.

**The consequences recur on the same tasks.** We split the 20 nonce blocks into two halves by an outcome-independent hash of the block name, ranked tasks by turnover on one half and measured the other half (Appendix Table 10). The 17 most fragile tasks (top 20%) carry 41.0% (crossed [26.6, 55.9]) and 42.3% ([27.0, 57.5]) of the held-out turnover in the two directions, against 20% if fragility did not persist. Their turnover is 38.7%, while the other 68 tasks still turn over 13.7%. Both groups have rescues and damages; neither group's net gain is resolved under the crossed interval. This replicates fragility on other nonces of the same tasks; it does not predict which new tasks are fragile.

**Split-half fragility (Table 10).** The halves are assigned by an outcome-independent hash of the block name (sha256 of a fixed prefix and the block name, mod 2), written to a file before this post hoc analysis loaded outcomes; fragility is a task's mean turnover over the four W4 cells and the defining half's blocks, and ties at the top-20% threshold are broken by a hash. In the reverse direction (rank on the second half, estimate on the first), the 17 fragile tasks turn over 39.0% ($t$ [34.2, 43.9]) and carry 42.3% of the held-out turnover, the other tasks 13.3%; the fragile tasks' net gain is $+3.7$ pp ($t$ [$+0.6$, $+6.9$]; crossed [$-5.0$, $+12.4$]) and the rest's $+0.5$. Ranking tasks by BF16 success instead (tertiles on one half) also separates turnover: the hardest tertile turns over 6.7–10.0% and the others 25.4–27.3%, with no net gain resolved under the crossed interval.

## C Telecom Holdout: Protocol, Readings and Replay Audit (supports §4)

This appendix supports the holdout statement of §4: the net estimate is negative, the frozen readings classify it as inconclusive because they had no negative category, and both pre-registered secondary checks passed.

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside $\pm 3$ pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical.

**Replay acceptance on the holdout (post hoc audit).** $Y_{\mathrm{QQ}}$ is the one arm taken from the original episode rather than re-executed, and an exploratory count showed only-QQ successes more often than random labelling would give in two q8 holdout cells. We therefore audited the holdout's acceptance records after the analysis, with a script that outputs only consistency checks and counts. In the initial audit, the fixed hash rule selected 135 of the 1202 divergent states for the QQ no-op; all 135 were run and passed, none was missing, failed or non-computable, and the remaining states had not yet been re-run. Beyond the frozen checks (suffix, prompts, termination, outcome, budget counters), the 135 also match on reward, database state, the official reward, the final environment hash against a strict replay of the original episode, and the history prefix. The original QQ and the re-run arms are scored by the same official reward rule: recomputing it from the stored reward information reproduces the stored value for every arm. The audit read the same input files as the frozen analysis, so no repair under the protocol's missing-data rules was needed, and the preregistered result stands as reported. A later full replay, under a protocol frozen before any task ran, re-ran QQ on all 1202 divergent states, with each task's Q checkpoint hash, serving-configuration hash and numerics canary checked equal to the original fork's at server start: all 1202 matched on every check above, none was missing or failed, and the original 135 matched again. The replays therefore rule out a replay artifact in the shared QQ baseline; they do not explain the only-QQ pattern, which a real upgrade that damages the original policy would also produce. Re-including the two structurally excluded task–cell pairs in all twelve blocks (post hoc sensitivity, frozen estimator) with every unit set to $-1$ or $+1$ gives point estimates of $G_{\mathrm{all}}$ from $-6.68$ [$-10.35$, $-3.01$] to $-2.27$ [$-5.96$, $+1.43$] pp (crossed 90%). This is a range of point estimates: the interval of the most optimistic scenario includes zero. With their observed units kept in the eleven blocks where they exist and only the two missing units set to $-1$ or $+1$, the estimates are $-4.72$ to $-4.35$ pp, each with a crossed 90% interval below zero. The frozen reading is inconclusive in all 36 combinations. Table 11 gives the confirmatory result. Table 12 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is $\leq 0$, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues ($-9.09$ pp) and the fewest-faults tercile ($-6.37$ pp).

### C.1 Telecom development tasks

On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\mathrm{all}} = +2.7$ [$-4.7$, $+10.1$] (GPTQ-W4) and $-0.6$ [$-4.9$, $+3.7$] pp (AWQ-style W4), against $-5.6$ and $-6.6$ on the holdout; q4 has no telecom development forks.

## D Reference Experiments in Full: Q′, Common States and Key Order (supports §5)

This appendix supports §5: two-sided flips and continuation dependence also appear without more bits; at common states B's extra action value over Q′ is not resolved and equivalence is not shown; and on the eligible states one key-order change that keeps the tool operation identical also changes outcomes (exploratory).

### D.1 Second same-bit-width checkpoint

Per model (final analysis set, $n_{\mathrm{blk}} = 6$), the turnover difference is $\Delta M = +0.6$ [$-3.1$, $+4.3$] pp in q4 and $-2.5$ [$-5.6$, $+0.5$] in q8 (fixed pool, 95%), with crossed intervals [$-5.5$, $+6.9$] and [$-7.3$, $+2.7$]. Within $\pm 3$ pp, equivalence is shown for no per-model $\Delta M$ or $\Delta G_{\mathrm{all}}$; the q8 off-diagonal difference ($+0.40$, 90% [$-2.12$, $+2.92$]) is the only per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate.

**Continuation dependence and the intervals drawn in Figure 7a,b.** On their own divergent states (784 per design), the swap's effect depends on the continuation in 26.2% of Q′'s divergent states (fixed pool [20.7, 31.3]), against 25.0% for B ([20.6, 29.0]); each same-precision rate (rescues, damages, continuation dependence) has a fixed-pool 95% lower bound above zero (Amendment A1.3). Fixed-pool 95% intervals drawn in Figure 4a, W4 (B; Q′): rescues [6.0, 10.1]; [7.0, 11.7]; damages [6.5, 11.0]; [6.4, 10.5]; turnover [13.9, 19.6]; [15.7, 19.8]. New-task (crossed) 95% intervals drawn in Figure 7a,b, W4 (B; Q′): rescues [5.0, 11.9]; [5.9, 13.2]; damages [5.7, 12.4];

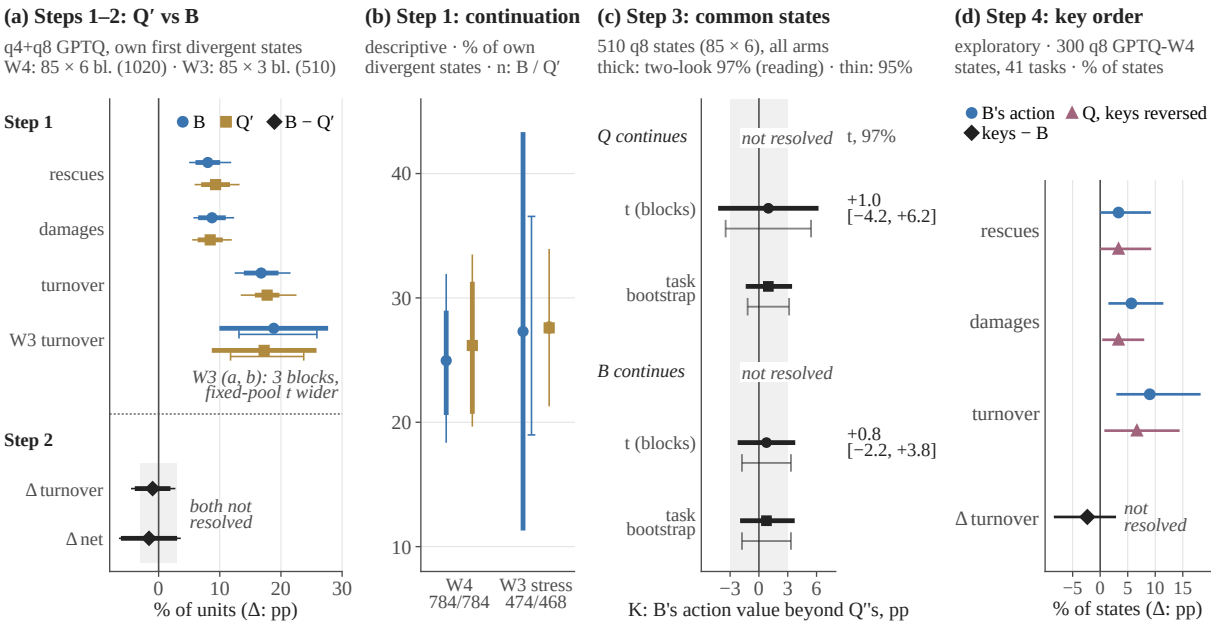



Figure 7: Complete reference display retained from the original analysis: (a) same-precision W4 and W3 effects and paired differences; (b) continuation-dependent shares; (c) common-state two-look and unadjusted intervals; (d) key-order effects. Thick/thin: fixed-pool/new-task 95% in (a,b), two-look 97%/unadjusted 95% in (c). Panel (d) uses the wider 95% interval. Populations remain separate.

[5.5, 12.0]; turnover [12.5, 21.6]; [13.4, 22.5]; continuation dependence [18.4, 31.9]; [19.7, 33.5]. W3 stress (474 and 468 divergent states; B; Q′): turnover fixed pool [9.9, 27.7]; [8.7, 25.8], crossed [13.1, 25.9]; [11.8, 23.7]; continuation dependence fixed pool [11.3, 43.3]; [27.1, 28.1], crossed [19.0, 36.6]; [21.3, 33.9]. With three blocks the fixed-pool $t$ interval can be wider than the crossed one.

Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 242/242 (same precision), and no directory was quarantined. At the unit level, Cohen's $\kappa$ between the action-side flip indicators of the two substitutions is 0.43 in q4 and 0.61 in q8; at the task level, cross-design agreement at matched noise is 0.765 of within-design replication.

W3 stress check (§5; $n_{\text{blk}} = 3$): every state passed Q replay, QQ no-op samples passed 31/31 and 39/39, and no directory was quarantined. All branches passed the prefix and environment checks except one of 774 in q8 same precision: a Q′Q′ branch that ended in an agent-side context overflow, for which the harness records no prefix comparison; its environment hash and first replayed prompt match the frozen state, and by the protocol it counts as a failure, not an exclusion.

**Equivalence tests (W4).** The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI $[-3.26, +1.30]$). No precision-specific excess was detected, but equivalence within $\pm 3$ pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\text{all}} = -1.57$ $[-5.17, +2.04]$, $p = 0.230$) and, descriptively, for the off-diagonal share ($-1.210$ $[-4.09, +1.67]$, $p = 0.133$, as the mean of per-block ratios, which the interval and test use; $-1.216$ as the ratio of block sums).

**Units affected by both, or by only one, substitution.** At W4, 107 units flip under both substitutions, 64 only under B and 74 only under the second checkpoint. A within-task block shuffle, which keeps each task's flip propensity, gives 76 [69, 84] units flipping under both, so the two substitutions share affected units beyond what flip-prone tasks explain. The overlap lies in action replacement: over the 1020 W4 units, Cohen's $\kappa$ between the indicators "B's action rescues" and "the second checkpoint's action rescues" is $\kappa_r = 0.60$, and for damages $\kappa_d = 0.54$. Under continuation-only takeover, the affected units are largely substitution-specific: rescue sets under the continuation-only contrast agree with $\kappa =$

0.20, and 70% of continuation-dependent units are continuation-dependent under one substitution only. The difference in action-side net value ($\Delta G_{\text{all}}$) comes only from the substitution-specific units and is not resolved ($-1.6$ pp; fixed pool $[-6.2, +3.0]$, new tasks $[-6.5, +3.6]$).

**Full substitution (W4).** Under full substitution B flips more units than Q′ (193 vs. 149; $+4.3$ $[+0.4, +8.3]$ pp, one of several contrasts); the action-side net difference ($\Delta G_{\text{all}}$, B's action against Q′'s under Q's continuation, not an $F$ difference) comes only from substitution-specific units and is not resolved ($-1.6$ $[-6.2, +3.0]$ pp). Only value measures on the same units reveal this structure.

**W3 stress pair.** The protocol's complete W3 panel (Q GPTQ-W3, Q′ a second W3 calibration draw; q4 and q8, $n_{\text{blk}} = 3$, blocks q3–q5, 510 units) is reported next to W4, never pooled with it. The same-precision substitution produces 17.3 pp of turnover, the BF16 substitution 18.8 pp; off-diagonal shares are 27.6% and 27.3%. $\Delta M = +1.6$ pp (95% $[-0.66, +3.80]$) is unresolved: the preregistered $\pm 3$ pp equivalence test fails ($p = 0.055$); that the 90% interval lies just above zero does not change this. $\Delta G_{\text{all}}$ is equivalent within $\pm 3$ pp only in the preregistered pooled population ($+1.2$, $p = 0.017$); per model it is $+3.1$ (q4) and $-0.8$ pp (q8). The off-diagonal difference is $-0.288$ (ratio of block sums) or $-0.307$ (mean of per-block ratios, used by the interval and test: 90% $[-11.0, +10.4]$, $p = 0.27$). On the action side ($G$), rescued and harmed task×block units overlap ($\kappa = 0.60, 0.68$); the designs may first diverge at different decisions, so this is unit overlap, not a same-state replication. Of 122 units whose $G$ outcome flips, 62 flip under both substitutions, 34 only under BF16 and 26 only under the other W3 checkpoint: 49% respond to only one. Flip volume, affected units and net upgrade value must be measured separately.

### D.2 Common states: extra value of the BF16 action

**Common-state comparison: details.** Above, each design selects its own divergent states. A comparison pre-registered at three blocks removes that difference: on 510 ordinary q8 states (85 tasks × 6 blocks, sampled by a fixed rule), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\text{B}, c) - V(\text{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both intervals exclude zero (Holm), equivalence only if both lie within $\pm 3$ pp (two one-sided tests). The design first ran three blocks, and that first look was unresolved (under Q's continuation $K = -0.4$ pp, 95%: $t$ $[-9.3, +8.5]$, bootstrap $[-3.5, +2.7]$; under B's $+2.4$ pp, $[-3.5, +8.2]$, $[-0.8, +5.5]$). Three more blocks were then added. That decision followed the unresolved first look: a dated amendment fixed the extension and the two-look rule after that look and before any extension run had produced output, and the protocol designates this two-look reading (Pocock levels from a normal approximation: about 97% intervals for a difference and 94% for equivalence) as the confirmatory one. The Pocock constant is not calibrated for small-$k$ $t$ intervals, the task bootstrap or the joint rule, so the two-look reading carries no exact error guarantee. Because the extension was chosen after the first look, even the two-look reading cannot fully protect the error rate conditional on the observed first look (protocol D2.2). We therefore print it next to the unadjusted reading at the frozen levels and claim neither an advantage nor equivalence unless both agree. After the fixed extension (six blocks), B's action is not shown to be worth more than Q′'s, and equivalence is not established. Under Q's continuation, $K = +1.0$ pp (95%: $t$ $[-3.5, +5.4]$, bootstrap $[-1.2, +3.1]$; two-look 97%: $[-4.2, +6.2]$, $[-1.4, +3.5]$). Under B's, $K = +0.8$ pp (95%: $[-1.8, +3.3]$ for both; two-look 97%: $[-2.2, +3.8]$, $[-2.0, +3.7]$). For equivalence under B's continuation, the ordinary 90% intervals lie within $\pm 3$ pp ($t$ $[-1.2, +2.8]$, bootstrap $[-1.4, +2.9]$) and the 94% intervals of the protocol-designated two-look confirmatory reading do not ($t$ $[-1.6, +3.2]$); because the extension followed the first look, we do not claim equivalence. Without the 26 states at which earlier studies had already produced outcomes, $K$ is $+0.5$ and $+0.8$ pp, also not resolved, and the unadjusted reading no longer shows equivalence under B's continuation either; BF16's own outcome at the extension states was also known in aggregate before the amendment (protocol D2.1). Under Q's continuation, B's action flips $M = 7.1\%$ of outcomes (rescue 4.1, damage 2.9) for a net $\Delta = +1.2$ pp (95%: $t$ $[-1.3, +3.6]$); Q′'s flips 8.0% (4.1, 3.9) for $+0.2$ pp ($[-2.3, +2.7]$; within $\pm 3$ pp by two one-sided tests). A second W4 checkpoint's ac-

tion departs from Q's at least as often as B's does (agreement with Q's action 53.9% against 55.7%; B with Q′ 57.3%). The interaction of $K$ with the continuation is $-0.2$ pp (95%: $t$ [$-4.5$, $+4.1$], bootstrap [$-2.7$, $+2.4$]; secondary, not resolved). *q4, a separate prospective analysis.* The same design on Qwen3-4B planned three blocks. The frozen quarantine rule removed f01 (below), leaving 170 states. $K = -1.2$ pp under Q's continuation (bootstrap 95% [$-5.9$, $+3.5$]) and $-2.4$ pp under B's ([$-6.5$, $+1.2$]), both not resolved. With two blocks the $t$ interval has one degree of freedom. Under Q's continuation the two block means happen to coincide, so it collapses to a point. That is a coincidence of two values, not a precise measurement or zero variance, and the reading rests on the bootstrap. *Identity checks.* Across both models, 1,238 of 1,240 reference and implicit identity checks passed. Both failures are one q4 state. There the substituted action was semantically identical to the original after parsing, but its token sequence differed in the order of keys inside a tool-call argument. The canonical comparison treats such actions as equal, but the chat template renders the order, so the next prompt differed and the continuation could not be a replay. Every check with a token-identical action passed (869/869). As the frozen rule requires, the block was removed. Changing outcomes is thus not improving them; this does not make precision irrelevant or the two actions equivalent.

**Common states, BF16 against Q′, by component.** At the same states and continuation, BF16's extra rescues over Q′ on q8 are $+0.3$ pp and its extra damages $-0.6$ pp, with the net $K = +0.9$ unresolved; each component lies within $\pm 3$ pp under a 90% two one-sided test (exploratory, post hoc; the margin is wide relative to rescue rates of 3–5%). q4 is unresolved.

### D.3 Key-order control

**Design.** Protocol `KEYORDER_PERTURB_PROTOCOL.md` (ko-v1, exploratory control). Eligible states are first tool-structure divergences of q8 GPTQ-W4 (Bank P) where Q's action is a tool call with at least two top-level argument keys, reversing them leaves the parsed call and the environment effects unchanged, and the next prompt tokens differ (300 of 1473 divergent states, 41 tasks). The arm KQ forces the reversed action and lets Q continue; QQ and BQ are reused from Bank P. Eligibility was fixed without outcomes; the analysis code (sha256-recorded) was frozen before any KQ outcome was read. All four tasks passed acceptance (KQ checks 300/300, shadow 300/300, QQ no-op 35/35, no errors or context overflows). Two tasks were restarted with resume during the run (Amendment 2026-10-05g).

**Results.** Conditional on eligible states (pooled ratio; intervals are the wider of block and crossed bootstrap), KQ changes 6.7% [0.8, 14.4] of outcomes and B 9.0% [2.9, 18.2]; KQ has 10 rescues and 10 damages, B 10 and 17, and the flips coincide on 5 rescues and 7 damages. Rescue and damage shares (Figure 4b): KQ 3.3% [0.0, 9.3] and 3.3% [0.4, 8.0]; B 3.3% [0.0, 9.2] and 5.7% [1.5, 11.5]. Over the full 1700-unit pool, KQ contributes 1.2 pp [0.1, 2.8] of turnover against 17.9 pp for the BF16 swap on all divergent states.

## E Trigger, Oracle and KL Detail (supports §6)

This appendix supports §6: the frozen Q-only trigger enriches outcome-changing decisions but does not select net-beneficial upgrades, and token KL tracks how often actions differ, not net value (exploratory, post hoc).

### E.1 The frozen trigger and its headroom

Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell × state rows, 58 flips), teacher-forcing both models on the generated ids of $a_{\mathrm{B}}$ and $a_{\mathrm{Q}}$. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

**Uncertainty is associated with divergence.** Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified). Among already divergent U1 states, no Q-visible statistic has an interval excluding AUC 0.5; small-subset associations vary across state sources. This development diagnosis motivated the frozen margin rule, whose validation reuses the tasks with fresh nonce blocks. On the secondary validation population, ordinary states from BF16 trajectories (U2B;

f02/f03, 1020 rows, 75 outcome-changing states), the frozen rule captures 4, 11 and 35 outcome-changing states against 3.81, 8.35 and 16.07 expected at random ($p = .5404$ and .1995 at 5 and 10%; the 20% excess is resolved), with net gains of $-2$, $-1$ and $+5$ against random $+0.07$, $+1.10$ and $+2.48$.

### E.2 Matched references for the frozen trigger

Table 14 compares the frozen trigger of §6 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per $a_{\mathrm{Q}}$ token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and $a_{\mathrm{Q}}$ at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations over the $a_{\mathrm{Q}}$ tokens were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; the maximum was chosen, and all of f02/f03 is reported. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

### E.3 Fidelity, sensitivity and signed value

**KL tracks action divergence and turnover, not net value.** On ordinary BF16-sourced states shared by every method (U2B), we compare methods by $D_{cc} = Y_{\mathrm{QB}} - Y_{\mathrm{BB}}$, Q's action against B's under the shared BF16 continuation, with turnover $M_{cc}$ (Appendix F.5). These comparisons evaluate complete quantization configurations, including their calibration pipelines, rather than isolated algorithmic components. Mean teacher-forced approximate token $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ tracks how often actions differ, not the direction of their consequences (correlations below), consistent with KL measuring the size of a disagreement rather than its direction (Nikolić et al., 2026). SpinQuant-W4A4 has lower KL than QuaRot-W4A4 (0.165 against 0.199), yet their pre-registered task-value contrast is unresolved ($+1.2$ pp, crossed $[-4.7, +7.1]$; Table 15).

The following decomposition was added after panel results were read and is exploratory. On each cell's admitted states, exact action equality and deterministic continuation give

$$M_{cc} = P(a_{\mathrm{Q}} \neq a_{\mathrm{B}})\, P(Y_{\mathrm{QB}} \neq Y_{\mathrm{BB}} \mid a_{\mathrm{Q}} \neq a_{\mathrm{B}}).$$

B's rescues $r_{cc} = P(Y_{\mathrm{BB}} = 1, Y_{\mathrm{QB}} = 0)$ and damages $d_{cc} = P(Y_{\mathrm{BB}} = 0, Y_{\mathrm{QB}} = 1)$ give $D_{cc} = d_{cc} - r_{cc}$. The decomposition separates how often an action changes from what a changed action does; it is diagnostic rather than a mechanism.

**Approximate KL.** Both models are teacher-forced on identical prompt ids and B's action tokens. At each token, $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ uses the union of both top-20 sets and the forced token, plus one tail bucket per model. An in-union token missing from a model's top-20 receives min(the 20th probability, tail mass divided by the number of missing tokens). The state score averages over B's action tokens. This coarse distribution is not the full token distribution.

**Across and within configurations.** Across eleven q8 cells, KL's descriptive Spearman correlation is 0.99 with the action-difference rate, 0.925 with $M_{cc}$, 0.81 with rescues and 0.93 with damages. It is 0.46 with the flip rate conditional on a difference and $-0.44$ with $D_{cc}$. Removing the three unquantized path controls (eight configurations) gives 0.98, 0.64 and $-0.52$ for the action-difference rate, conditional flip rate and $D_{cc}$. These configurations are not exchangeable draws. Within cells, the mean AUC is 0.93 for an action difference, 0.46 for a flip conditional on a difference and 0.55 for a rescue conditional on a flip; the last uses 3–39 flips per cell. These diagnostics establish no reliable predictor of net upgrade value.

## F Scope Checks: Ordinary States, Other Environments, a Second Family and Other Configurations (supports Table 1)

This appendix supports Table 1: it reports each scope check in full and states where a check does not support extending the main-panel statements.

### F.1 Ordinary states and the user simulator

**Ordinary states turn over less, mostly through text under our simulator.** At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; $t$ intervals over block means), $M$ per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within $\pm 2.8$ pp (U1, U2B) and $\pm 5.0$ pp (U2Q), and every interval covers zero (Appendix F.2). Identical actions never change the outcome, as determinism implies. Reused arms rest on canonical equality. All 4,114 such reuses were rendered with each model's chat template; 4,113 are token-identical. The remaining pair differs only in JSON key order, and its two cross arms, measured separately, gave the reused outcomes. After deduplication, 150 of the 208 outcome-changing cell×state rows ($\approx$72.1%) are class (ii), the same tool structure with different text; 26 involve a different tool name, call count or argument (Table 17). The rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more.

**Sensitivity to the user simulator.** Text actions act through the user simulator, and changing the simulator moves $\tau$-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels.

**Sensitivity of the four-branch results.** Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change $-6.0$ $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 6 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings.

### F.2 Bank U per Sub-Bank

Table 16 gives the per-state turnover behind Appendix F.1, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

### F.3 Models compared on their own ordinary states

**Mistral population, scoring and coverage.** The Mistral panel applies the same row rule to Mistral's own BF16 trajectories on the 85 retail and airline tasks. Seven BF16 source rows are missing because the source episode ended in an invalid tool call produced by the model itself (airline task 44 in all six blocks, airline task 13 in f01). A dated amendment, fixed from the BF16 source runs before any Mistral fork output existed, excludes both tasks in every block and cell, leaving 83 tasks. The selection was outcome-blind, but BF16 Mistral fails these tasks structurally, so the estimates describe this conditional population, not the 85-task panel, and are not compared with q8 as if the populations were equal. A branch that ended because the harness could not accept the agent's own message (an empty or invalid tool name, or an empty message) is scored as a failure, as in end-to-end evaluation. This rule was added after the Mistral phase-1 results had been read; without it the Mistral primary comparison is not estimable

from phase 1 (one complete block). The analysis without the rule (which leaves such an arm missing and fails its block) and one applying the rule only to data not yet read when it was fixed are reported alongside; both admit the same three blocks for GPTQ − RTN (+1.2 [−13.8, +16.2]). Coverage: GPTQ-W4 has 498 states on six blocks and RTN-W4 415 states on five. Full estimates on the 83 tasks (GPTQ-W4 on six blocks, RTN-W4 on five; Table 18): in GPTQ-W4, B's action rescues $r = 4.6$ [2.4, 6.8] and damages $d = 5.4$ [2.4, 8.5] pp of states ($M = 10.0$ [7.0, 13.1]; fixed-pool 95%), and no mode's net value is resolved ($G = -0.8$ [−5.2, +3.6], $H = +1.4$ [−3.0, +5.8], $F = +0.0$ [−4.2, +4.2]). RTN-W4 is similar ($M = 10.4$ [6.3, 14.4]). The pre-registered contrast GPTQ − RTN is not resolved: $\Delta D_{cc} = +2.2$ [−3.5, +7.8] pp, $p = .35$, on five blocks, two of which are admitted only by a scoring rule adopted after an interim look (above).

### F.4 Replacement and four-branch estimands on BFCL and ALFWorld (supplementary)

Table 19 gives the replacement quantities of §4 ($r$, $d$, $M$ and $G_{\text{all}}$ over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The four-branch columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{\text{QQ}} = 0$ on every state, so $\Delta_{\text{Q}}$ is almost constant and their off-diagonal mass is $H$. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%). The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ [−43.5, −25.4] pp; KV4 cells reduce to $H$.

### F.5 Quantization methods on shared ordinary states

These tests ask whether numerical error locates changed actions, consequential changes and beneficial changes. Methods share ordinary BF16-sourced states (U2B); the separate Mistral population is specified in Appendix F.3.

**Contrasts.** Methods are compared on shared states by $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ ($= -\Delta_{\text{B}}$): Q's action against B's under the common BF16 continuation. Because BB is the shared BF16 original, two methods differ on the same states only through $Y_{\text{QB}}$. One pre-registered Holm family holds three contrasts on $D_{cc}$, with $t$ intervals over block-mean differences: P1, QuaRot-W4 − GPTQ-W4; P2, SpinQuant-W4A4 − QuaRot-W4A4; P3, KIVI-2 − KV8. Unless marked crossed, intervals for these panels in §6 and §7 are fixed-pool 95% ($t$ over block means, as fewer than 10 blocks are complete); other new-task (crossed) intervals are in the analysis outputs.

**Numerical quality.** QuaRot-W4 against GPTQ-W4 is −0.4 [−10.7, +9.9] pp (P1). The matched SpinQuant-W4A4 contrast is P2 (complete family: Table 20; paired diagnostic components: Table 15). Its third block rests on a one-time re-run of arms lost to a user-simulator overload; on the two blocks without it the contrast is +2.4 pp (strict view).

**Upgrade modes on the q8 panel.** On the q8 ordinary-state panel, B's action alone ($G$) is resolved in no cell. Fixed-pool intervals for B's takeover $H$ or full substitution $F$ exclude zero in weight-only (AWQ-style W4, QuaRot-W4), W4A4 and KV8 cells, and new-task intervals still do in both W4A4 cells and KV8 (unadjusted), so resolved value is not confined to quantized activations. It is largest in SpinQuant-W4A4 ($H = +11.8$ [6.7, 16.8], $F = +13.7$ [9.3, 18.2] pp).

**KIVI.** KIVI (Liu et al., 2024) quantizes keys per channel and values per token. KIVI-2 against KV8 is not resolved (−0.6 [−38.0, +36.8] pp), nor is KIVI-4 against KV8 (+1.6 [−2.9, +6.0] pp). Our implementation is a replay-compatible variant: at every decode step it reconstructs the quantized view that official KIVI's token-by-token decode would hold at that position from the full-precision cache, rather than persisting a compressed cache across turns, so every token, prompt or generated, attends with that view. The KIVI-2 contrast (P3) rests on two blocks: the third lost one arm to a server timeout that recurred on a single-request re-run, and both admitted blocks depend on the re-run of arms lost to server failures (strict rule: not estimable). Net of the kernel path, the quantization effect is +0.6 [−51.7, +52.9] (KIVI-2) and +2.4 [−5.4, +10.1] pp (KIVI-4), and the unquantized

kernel path itself has $D_{cc} = -1.2$ $[-6.2, +3.9]$.

**All admitted ordinary-state upgrade effects.** Table 21 reports all three upgrade modes for every admitted configuration, with both fixed-pool and new-task intervals.

## G Provenance of the Retained Evidence

Figure values, retained tables and scoring definitions map to committed analysis outputs and frozen manifests. The supplement retains the complete primary contrasts, inference rules, acceptance checks and dated protocol amendments needed to assess the main claims. Additional research branches are preserved in the project history but are not part of this paper's evidence chain.

| Config. (models) | Method, weights | Group, grid | Act-order | Calibration | KV cache / activations / rotation |
|---|---|---|---|---|---|
| GPTQ-W4 Q (q4, q8) | GPTQ, W4; damp 0.01, block 128, true-sequential | 128, asym. | yes | draw c1: 57 BF16 agent-trajectory states (39 retail, 18 airline calibration tasks, disjoint from the 85 dev tasks), one state per task, full context (median 6041 / 6310 tok.); 64,000 activation rows (1122–1123 per state); draw seed 20260729 | KV BF16, act. BF16, no rotation |
| GPTQ-W4 Q′ (q4, q8) | as Q | 128, asym. | yes | draw c2, seed 20260730: same 57 tasks and trajectories, re-drawn decision state and rows (5/57 states shared with c1); not disjoint | as Q |
| AWQ-style W4 (q4, q8) | AWQ-style: per-linear scale $\lvert\bar{x}\rvert^{\alpha}$, 20-point $\alpha$ grid, scale folded into weight, then RTN; no clip search | 128, asym. | n/a (GPTQ only) | draw c1 as GPTQ-W4 Q; $\alpha$ fit on a 4096-row sample (code default; per-run value not recorded) | as Q |
| GPTQ-W3$^{s}$ Q / Q′ (q4, q8) | GPTQ, W3 | 128, asym. | yes | c1 / c2 as W4 | as Q |
| RTN-W3$^{s}$ (q4, q8) | RTN, W3 | 128, asym. | n/a | none | as Q |
| KV8 (q4, q8) | BF16 weights | – | – | none | INT8 K and V, asym. min–max per (token, KV head) over head_dim, no clip, no rotation; fake quant in fp32 |
| KV4-rot$^{s}$ (q4, q8) | BF16 weights | – | – | none | INT4 K and V, same grid as KV8; online normalized Hadamard on head_dim: q and k rotated, V rotated and rotated back |
| QuaRot-W4A4 (q8) | random-Hadamard rotation (seed 0) + GPTQ W4, MSE clip search | per out-channel, sym. | no | WikiText-2 train, 128 × 2048 tok., seed 0 | A4: sym. per-token dynamic, clip 0.9, inputs of qkv, o, gate/up, down, online Hadamard (o, down); KV16, per-head q/k Hadamard after RoPE |
| SpinQuant-W4A4 (q8) | learned $R_1$, per-layer $R_2$ (100 steps, lr 1.5, A4 in training) + GPTQ W4 as QuaRot | per out-channel, sym. | no | weights and rotations: WikiText-2, 128 × 2048, seed 0 | as QuaRot-W4A4 |
| KIVI-2 / KIVI-4 (q8) | BF16 weights | – | – | none | K per channel, V per token, group 32, asym.; last 128 tokens full precision |
| Mistral GPTQ-W4 / RTN-W4 (m24) | GPTQ W4 / RTN W4 | 128, asym. | yes / n/a | GPTQ: draw c1, 57 states, 64,000 rows, seed 20260729 (no c2 draw) | as Q |

Table 4: Quantization configuration of every checkpoint configuration, from the checkpoint metadata. AWQ-style W4 is AWQ-style (scale folded into the weight, then RTN, no clip search). Q′ re-draws the decision state and activation rows from the same 57 calibration trajectories as Q. The W4A4 configurations keep a 16-bit KV cache and calibrate on WikiText-2, so a QuaRot-minus-GPTQ contrast also changes the calibration source. One q8 calibration context (c1) is marked truncated. All checkpoints are fake-quantized and stored in BF16; embeddings, `lm_head` and norms stay BF16. $^{s}$stress cell.

| Cell | units | div. | $r$ | $d$ | $M$ | $G_{\text{all}}$ [fixed pool] | [new tasks] |
|---|---|---|---|---|---|---|---|
| q4 GPTQ-W4 | 85×20 | 1102 | 6.8 | 8.7 | 15.5 | $-1.9$ [$-3.7$, $-0.1$] | [$-6.8$, $+2.7$] |
| q4 AWQ-style W4 | 85×20 | 1439 | 9.8 | 10.1 | 19.9 | $-0.3$ [$-2.2$, $+1.6$] | [$-5.4$, $+5.2$] |
| q8 GPTQ-W4 | 85×20 | 1473 | 10.4 | 7.5 | 17.9 | $+2.9$ [$+1.4$, $+4.4$] | [$-2.9$, $+9.2$] |
| q8 AWQ-style W4 | 85×20 | 1562 | 11.4 | 9.5 | 20.9 | $+1.9$ [$-0.6$, $+4.3$] | [$-3.9$, $+8.1$] |
| W4 pool | 85×20 | | 9.6 | 9.0 | 18.6 | $+0.7$ [$-0.3$, $+1.6$] | [$-2.1$, $+3.4$] |
| q4 KV8 | 85×22 | 593 | 3.1 | 3.4 | 6.5 | $-0.3$ [$-1.7$, $+1.0$] | [$-3.0$, $+2.3$] |
| q8 KV8 | 85×22 | 855 | 4.3 | 3.7 | 8.1 | $+0.6$ [$-0.4$, $+1.6$] | [$-1.9$, $+3.0$] |
| q4 GPTQ-W3$^s$ | 85×3 | 232 | 9.0 | 10.2 | 19.2 | $-1.2$ [$-7.0$, $+4.7$] | [$-8.6$, $+5.9$] |
| q4 KV4-rot$^s$ | 85×3 | 233 | 11.8 | 7.1 | 18.8 | $+4.7$ [$-4.1$, $+13.5$] | [$-3.1$, $+12.9$] |
| q8 GPTQ-W3$^s$ | 85×3 | 242 | 10.2 | 8.2 | 18.4 | $+2.0$ [$-12.5$, $+16.4$] | [$-6.3$, $+11.8$] |
| q8 KV4-rot$^s$ | 85×3 | 239 | 11.8 | 8.6 | 20.4 | $+3.1$ [$-14.5$, $+20.8$] | [$-5.9$, $+12.9$] |
| q8 RTN-W3$^s$ | 85×3 | 255 | 3.1 | 3.5 | 6.7 | $-0.4$ [$-3.8$, $+3.0$] | [$-4.7$, $+3.9$] |

Table 5: Replacing Q's first tool-structure-divergent action while Q continues, $\tau^2$-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix A.2). Fixed pool: block bootstrap ($\geq$10 blocks) or $t$ over block means; new tasks: crossed task×block bootstrap. $^s$stress cell, 3 blocks.

**A. Gains and action–continuation comparison**

| Cell | states | $G$ (this step) | $H$ (next step on) | $F$ (both) | $G - H$ |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | 1102 | $-2.9$ [$-5.7$, $-0.2$] | $+0.7$ [$-1.6$, $+3.0$] | $-2.5$ [$-5.3$, $+0.2$] | $-3.6$ [$-6.6$, $-0.6$] |
| q4 AWQ-style W4 | 1439 | $-0.3$ [$-2.6$, $+1.9$] | $+0.9$ [$-1.1$, $+2.9$] | $-0.8$ [$-3.3$, $+1.6$] | $-1.3$ [$-3.0$, $+0.6$] |
| q8 GPTQ-W4 | 1473 | $+3.3$ [$+1.6$, $+5.0$] | $+0.4$ [$-1.2$, $+1.9$] | $+5.4$ [$+3.5$, $+7.2$] | $+2.9$ [$+1.3$, $+4.7$] |
| q8 AWQ-style W4 | 1562 | $+2.1$ [$-0.6$, $+4.7$] | $-1.3$ [$-3.5$, $+0.9$] | $+0.6$ [$-1.5$, $+2.7$] | $+3.4$ [$+1.5$, $+5.2$] |
| W4 pool | 5576 | $+0.5$ [$-0.7$, $+1.7$] | $+0.2$ [$-1.3$, $+1.6$] | $+0.7$ [$-0.6$, $+2.1$] | $+0.4$ [$-0.8$, $+1.7$] |
| q4 KV8 | 593 | $-1.0$ [$-5.3$, $+3.2$] | $-2.5$ [$-4.5$, $-0.5$] | $-1.2$ [$-4.7$, $+2.3$] | $+1.5$ [$-3.1$, $+6.0$] |
| q8 KV8 | 855 | $+1.3$ [$-0.9$, $+3.4$] | $-0.4$ [$-2.1$, $+1.5$] | $+2.5$ [$+0.0$, $+5.1$] | $+1.6$ [$-0.6$, $+3.9$] |
| KV8 pool | 1448 | $+0.1$ [$-2.8$, $+3.0$] | $-1.4$ [$-2.7$, $-0.1$] | $+0.6$ [$-1.8$, $+3.1$] | $+1.6$ [$-1.2$, $+4.1$] |
| q4 GPTQ-W3$^s$ | 232 | $-1.3$ [$-7.7$, $+5.0$] | $+4.3$ [$-10.7$, $+19.4$] | $+1.3$ [$-1.9$, $+4.4$] | $-5.6$ [$-15.0$, $+3.7$] |
| q4 KV4-rot$^s$ | 233 | $+5.2$ [$-4.5$, $+14.9$] | $+9.4$ [$+5.1$, $+13.7$] | $+12.4$ [$+11.0$, $+13.9$] | $-4.3$ [$-18.3$, $+9.8$] |
| q8 GPTQ-W3$^s$ | 242 | $+2.1$ [$-13.3$, $+17.2$] | $+0.4$ [$-7.4$, $+8.2$] | $+2.1$ [$-7.4$, $+11.4$] | $+1.7$ [$-9.1$, $+12.3$] |
| q8 KV4-rot$^s$ | 239 | $+3.3$ [$-15.4$, $+21.9$] | $-0.8$ [$-18.2$, $+16.3$] | $+4.6$ [$-16.0$, $+25.0$] | $+4.2$ [$+2.5$, $+5.8$] |
| q8 RTN-W3$^s$ | 255 | $-0.4$ [$-3.8$, $+3.0$] | $+15.3$ [$+0.7$, $+29.9$] | $+18.4$ [$+16.7$, $+20.1$] | $-15.7$ [$-27.5$, $-3.9$] |

**B. Joint-upgrade contrasts and outcome patterns**

| Cell | $F - G$ | $F - H$ ($= \Delta_{\mathrm{B}}$) | off-diag. % | rev. % | mean $I$ |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | $+0.5$ [$-1.2$, $+2.2$] | $-3.2$ [$-6.4$, $+0.2$] | 25.0 | 2.6 | $+0.3$ |
| q4 AWQ-style W4 | $-0.4$ [$-2.4$, $+1.5$] | $-1.7$ [$-3.6$, $+0.4$] | 29.3 | 3.1 | $+1.3$ |
| q8 GPTQ-W4 | $+2.1$ [$+0.2$, $+4.0$] | $+5.0$ [$+3.2$, $+6.8$] | 22.9 | 2.7 | $-1.7$ |
| q8 AWQ-style W4 | $-1.5$ [$-4.2$, $+0.8$] | $+1.9$ [$-0.1$, $+3.8$] | 30.5 | 4.2 | $+0.3$ |
| W4 pool | $+0.2$ [$-1.0$, $+1.2$] | $+0.5$ [$-1.0$, $+2.1$] | 26.9 | 3.2 | $+0.0$ |
| q4 KV8 | $-0.2$ [$-3.5$, $+3.7$] | $+1.3$ [$-2.5$, $+4.9$] | 15.7 | 1.2 | $-2.4$ |
| q8 KV8 | $+1.2$ [$-0.5$, $+3.2$] | $+2.8$ [$+0.3$, $+5.5$] | 14.9 | 1.2 | $-1.5$ |
| KV8 pool | $+0.5$ [$-1.6$, $+2.8$] | $+2.1$ [$-0.2$, $+4.2$] | 15.3 | 1.2 | $-1.9$ |
| q4 GPTQ-W3$^s$ | $+2.6$ [$-0.6$, $+5.8$] | $-3.0$ [$-15.1$, $+9.1$] | 31.5 | 3.9 | $+1.7$ |
| q4 KV4-rot$^s$ | $+7.3$ [$-2.0$, $+16.5$] | $+3.0$ [$-1.9$, $+8.0$] | 33.0 | 1.7 | $+2.1$ |
| q8 GPTQ-W3$^s$ | $+0.0$ [$-11.0$, $+11.1$] | $+1.7$ [$-0.1$, $+3.4$] | 23.1 | 4.5 | $+0.4$ |
| q8 KV4-rot$^s$ | $+1.3$ [$-1.8$, $+4.3$] | $+5.4$ [$+0.8$, $+10.0$] | 31.0 | 4.6 | $-2.1$ |
| q8 RTN-W3$^s$ | $+18.8$ [$+13.8$, $+23.9$] | $+3.1$ [$-13.0$, $+19.2$] | 29.0 | 0.4 | $-3.5$ |

Table 6: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$, $H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}}$, $F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}}$. Off-diag.: share of states with $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$; rev.: strict reversals, $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$; mean $I$: mean action–continuation interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}} \in \{-2, \ldots, +2\}$. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; $t$ over 3 blocks for $^s$stress cells, which cannot be pooled). New-task intervals are in Table 7. Within a row, contrasts are paired and identified; between rows, descriptive. Intervals are not adjusted for multiplicity: the four core cells' $G - H$ form a Holm family ($m = 4$; q4 GPTQ-W4 Holm $p = 0.069$), their new-task intervals all cover zero, and $G - H > 0$ ranks two modes, not either against QQ.

**A. Upgrade gains**

| Cell | $G$ | $H$ | $F$ |
|---|---|---|---|
| q4 GPTQ-W4 | $[-10.3, +4.2]$ | $[-4.0, +6.0]$ | $[-9.9, +4.7]$ |
| q4 AWQ-style W4 | $[-6.5, +6.1]$ | $[-3.7, +5.5]$ | $[-7.4, +5.8]$ |
| q8 GPTQ-W4 | $[-3.3, +10.5]$ | $[-3.8, +4.9]$ | $[-2.0, +13.0]$ |
| q8 AWQ-style W4 | $[-4.1, +8.7]$ | $[-7.1, +4.6]$ | $[-6.2, +7.3]$ |
| W4 pool | $[-2.9, +3.8]$ | $[-2.7, +2.8]$ | $[-3.1, +4.4]$ |
| q4 KV8 | $[-9.4, +7.3]$ | $[-6.4, +1.1]$ | $[-9.3, +7.7]$ |
| q8 KV8 | $[-4.0, +6.9]$ | $[-3.5, +2.9]$ | $[-3.5, +8.1]$ |
| KV8 pool | $[-5.3, +5.6]$ | $[-3.7, +0.9]$ | $[-4.5, +5.8]$ |
| q4 GPTQ-W3$^s$ | $[-9.7, +6.6]$ | $[-6.6, +14.5]$ | $[-6.8, +9.4]$ |
| q4 KV4-rot$^s$ | $[-3.4, +14.2]$ | $[+2.1, +17.0]$ | $[+3.0, +21.6]$ |
| q8 GPTQ-W3$^s$ | $[-6.7, +12.1]$ | $[-7.2, +8.1]$ | $[-6.6, +11.8]$ |
| q8 KV4-rot$^s$ | $[-6.6, +13.8]$ | $[-11.2, +8.5]$ | $[-6.6, +15.9]$ |
| q8 RTN-W3$^s$ | $[-4.7, +3.9]$ | $[+5.9, +25.9]$ | $[+9.4, +28.2]$ |

**B. Paired differences between upgrade modes**

| Cell | $G - H$ | $F - G$ | $F - H$ |
|---|---|---|---|
| q4 GPTQ-W4 | $[-11.2, +3.6]$ | $[-4.7, +6.1]$ | $[-11.1, +4.4]$ |
| q4 AWQ-style W4 | $[-7.7, +5.1]$ | $[-6.2, +5.4]$ | $[-8.7, +4.7]$ |
| q8 GPTQ-W4 | $[-3.2, +9.3]$ | $[-3.0, +7.1]$ | $[-2.0, +12.5]$ |
| q8 AWQ-style W4 | $[-1.6, +8.6]$ | $[-7.6, +4.2]$ | $[-3.8, +7.5]$ |
| W4 pool | $[-3.1, +3.7]$ | $[-2.8, +3.1]$ | $[-3.5, +4.5]$ |
| q4 KV8 | $[-7.1, +9.7]$ | $[-5.7, +5.3]$ | $[-6.7, +9.5]$ |
| q8 KV8 | $[-3.8, +7.2]$ | $[-2.5, +5.0]$ | $[-3.1, +8.7]$ |
| KV8 pool | $[-3.8, +6.6]$ | $[-2.8, +3.9]$ | $[-3.0, +7.3]$ |
| q4 GPTQ-W3$^s$ | $[-15.0, +4.0]$ | $[-5.2, +10.5]$ | $[-12.2, +6.6]$ |
| q4 KV4-rot$^s$ | $[-13.5, +7.0]$ | $[-1.3, +17.1]$ | $[-6.2, +12.2]$ |
| q8 GPTQ-W3$^s$ | $[-7.7, +11.2]$ | $[-9.1, +9.1]$ | $[-6.8, +11.0]$ |
| q8 KV4-rot$^s$ | $[-3.3, +12.2]$ | $[-7.2, +9.9]$ | $[-2.5, +14.6]$ |
| q8 RTN-W3$^s$ | $[-24.7, -7.1]$ | $[+10.2, +27.8]$ | $[-7.1, +13.3]$ |

Table 7: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 6 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 6 treat the 85 tasks as fixed. $^s$stress cell, 3 blocks.

**A. Sensitivity to the action and continuation**

| Group | states | action changes | continuation changes | gap (pp) |
|---|---|---|---|---|
| $\tau^2$-bench W4 pool (4 cells) | 5576 | 34.9 [33.6, 36.0] | 29.6 [28.5, 30.7] | $+5.2$ [$+4.4$, $+6.1$] |
| $\tau^2$-bench KV8 pool | 1448 | 26.8 [25.0, 28.6] | 17.0 [14.9, 19.1] | $+9.8$ [$+8.0$, $+11.6$] |
| Telecom q8 W4 pool | 957 | 22.1 [18.2, 25.9] | 21.9 [17.8, 26.1] | $+0.1$ [$-2.1$, $+2.4$] |
| Same-precision W4 (Q′ as B) | 784 | 33.5 [29.8, 37.1] | 28.7 [24.8, 32.2] | $+4.9$ [$+2.9$, $+6.9$] |
| BF16, same blocks | 784 | 35.1 [32.6, 37.4] | 27.8 [23.8, 31.6] | $+7.3$ [$+4.9$, $+9.7$] |
| q8 RTN-W3[s] | 255 | 29.8 [28.1, 31.5] | 37.6 [26.0, 49.3] | $-7.8$ [$-18.1$, $+2.4$] |
| BFCL W4 pool[b] | 2296 | 18.4 [16.4, 20.5] | 11.2 [9.8, 12.6] | $+7.2$ [$+5.8$, $+8.7$] |
| ALFWorld q8 W4 pool[b] | 234 | 44.9 [38.3, 52.4] | 41.5 [34.3, 49.4] | $+3.4$ [$-1.8$, $+8.4$] |

**B. Outcome classes and named patterns**

| Group | unch. | act. only | cont. only | one arm | rev. | joint resc. | resc. undone |
|---|---|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool (4 cells) | 62.4 | 7.9 | 2.7 | 23.8 | 3.2 | 4.0 | 4.8 |
| $\tau^2$-bench KV8 pool | 71.5 | 11.6 | 1.7 | 14.1 | 1.2 | 3.0 | 2.3 |
| Telecom q8 W4 pool | 74.8 | 3.2 | 3.1 | 18.1 | 0.7 | 3.2 | 4.2 |
| Same-precision W4 (Q′ as B) | 64.0 | 7.4 | 2.5 | 22.5 | 3.6 | 2.3 | 5.8 |
| BF16, same blocks | 62.0 | 10.2 | 2.9 | 22.4 | 2.6 | 5.0 | 3.8 |
| q8 RTN-W3[s] | 61.6 | 0.8 | 8.6 | 28.6 | 0.4 | 12.6 | 0.8 |
| BFCL W4 pool[b] | 80.8 | 8.1 | 0.8 | 9.8 | 0.5 | 2.4 | 1.4 |
| ALFWorld q8 W4 pool[b] | 50.0 | 8.5 | 5.1 | 33.4 | 3.0 | 9.4 | 5.5 |

Table 8: Which intervention changes the outcome, in % of divergent states (pools: equal weight per cell). "Action changes": the outcome changes with the action under at least one continuation; "continuation changes": with the continuation controller under at least one action; gap = action only − continuation only. Fixed-pool 95% intervals: block bootstrap for the 20- and 22-block $\tau^2$-bench W4 and KV8 pools; $t$ over per-block ratios for the 6-block telecom and same-precision rows and the 3-block [s]stress cell; [b]one run per item, item bootstrap. Same-precision rows: Q′, a second GPTQ checkpoint of the same bit width, in the B role, and BF16 on the same blocks. New-task (crossed) intervals for the $\tau^2$-bench W4 pool: above.

**A. Joint rescue**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 3.96 | 3.48 | $+0.47$ [$+0.06$, $+0.87$] | [$-0.33$, $+1.37$] |
| q4 W4 cells | 2541 | 3.22 | 3.56 | $-0.34$ [$-0.83$, $+0.13$] | [$-1.46$, $+0.88$] |
| q8 W4 cells | 3035 | 4.69 | 3.40 | $+1.29$ [$+0.79$, $+1.78$] | [$-0.07$, $+2.74$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.99 | 2.17 | $+0.81$ [$+0.10$, $+1.52$] | [$-0.51$, $+2.19$] |
| Telecom q8 W4 pool | 957 | 3.24 | 3.94 | $-0.70$ [$-1.46$, $+0.05$] | [$-2.47$, $+1.15$] |
| Same-precision W4 (Q′ as B) | 784 | 2.27 | 3.50 | $-1.23$ [$-2.47$, $-0.04$] | [$-2.88$, $+0.51$] |
| BF16, same blocks | 784 | 4.96 | 3.53 | $+1.43$ [$-0.40$, $+3.42$] | [$-0.90$, $+4.26$] |
| Mistral, own BF16 states$^m$ | 464 | 3.4 | 2.9 | $+0.5$ [$-2.4$, $+3.5$] | [$-2.0$, $+3.6$] |

**B. Rescue undone**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 4.80 | 5.78 | $-0.99$ [$-1.58$, $-0.42$] | [$-2.19$, $+0.27$] |
| q4 W4 cells | 2541 | 4.77 | 6.01 | $-1.24$ [$-2.09$, $-0.45$] | [$-2.97$, $+0.69$] |
| q8 W4 cells | 3035 | 4.82 | 5.56 | $-0.74$ [$-1.35$, $-0.07$] | [$-2.34$, $+0.98$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.31 | 4.58 | $-2.28$ [$-2.98$, $-1.58$] | [$-3.62$, $-0.98$] |
| Telecom q8 W4 pool | 957 | 4.18 | 5.13 | $-0.95$ [$-1.70$, $-0.20$] | [$-2.63$, $+0.61$] |
| Same-precision W4 (Q′ as B) | 784 | 5.77 | 5.75 | $+0.02$ [$-2.33$, $+2.25$] | [$-2.66$, $+2.92$] |
| BF16, same blocks | 784 | 3.82 | 6.14 | $-2.32$ [$-4.46$, $-0.19$] | [$-4.68$, $+0.17$] |
| Mistral, own BF16 states$^m$ | 464 | 4.0 | 4.7 | $-0.7$ [$-2.0$, $+0.7$] | [$-3.0$, $+1.9$] |

Table 9: Joint rescue and rescue undone against the conditional label-permutation reference (keeps the number $k$ of successful branches per state), % of divergent states; exploratory, post hoc. 95% intervals: fixed pool (block bootstrap for the 20- and 22-block $\tau^2$-bench pools, $t$ over per-block ratios for fewer than 10 blocks: 6, or 5 for Mistral), new tasks (crossed task×block). Same-precision rows each fork at their own first divergence. $^m$States whose actions differ, on BF16 trajectories (BB is the BF16 original), five blocks; not comparable with Bank P rows.

| Tasks | | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| fragile | est. | 21.1 | 17.6 | 38.7 | $+3.4$ |
| (top 20%) | $t$ | [18.6, 23.5] | [14.3, 21.0] | [34.9, 42.6] | [$-1.0$, $+7.9$] |
| | crossed | [14.8, 28.1] | [12.4, 23.3] | [32.3, 45.7] | [$-7.1$, $+13.7$] |
| rest | est. | 6.5 | 7.2 | 13.7 | $-0.7$ |
| | $t$ | [5.2, 7.8] | [6.2, 8.3] | [12.5, 14.9] | [$-2.7$, $+1.2$] |
| | crossed | [4.4, 9.0] | [4.8, 9.9] | [10.2, 17.5] | [$-4.1$, $+2.4$] |

Table 10: Repeatable task fragility (exploratory, post hoc; four $\tau^2$-bench W4 cells, equal weight). Tasks are ranked by turnover on one hash-defined half of the 20 nonce blocks and measured on the other half (pp of units; 95% intervals, $t$ over held-out blocks and crossed task×block; direction A to B shown). A replication on other nonces of the same tasks, not a prediction for new tasks. Reverse direction: text.

| Cell (weight) | units | excl. | $G$ | $r$ | $d$ | $\rho_c$ |
|---|---|---|---|---|---|---|
| q4 GPTQ-W4 (1/2) | 396 | 1 | $-3.03$ [$-7.86$, $+1.80$] | $+5.05$ | $+8.08$ | 0.668 |
| q8 GPTQ-W4 (1/4) | 408 | 0 | $-5.64$ [$-9.33$, $-1.94$] | $+4.41$ | $+10.05$ | 0.374 |
| q8 AWQ-style W4 (1/4) | 396 | 1 | $-6.57$ [$-11.98$, $-1.15$] | $+6.31$ | $+12.88$ | 0.644 |
| Weighted ($G_{\text{all}}$) | | | $-4.57$ [$-7.17$, $-1.96$] | $+5.21$ | $+9.77$ | 0.588 |

Table 11: Telecom holdout, pre-registered analysis (34 tasks × 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional ($G_{\text{all}}$, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Alongside interval for the weighted row, $t$ over block means (not decisional): $-4.57$ [$-6.08$, $-3.05$].

| Family | Level | $n$ | $G$ [crossed 90%] | $p$ | Holm $p$ |
|---|---|---|---|---|---|
| step budget[a] | T1 | 138/129/110 | $-0.32$ [$-1.57$, $+0.94$] | .6771 | .6771 |
| | T2 | 124/126/162 | $-2.30$ [$-3.88$, $-0.72$] | .0165 | .0495 |
| | T3 | 123/144/124 | $-1.94$ [$-3.35$, $-0.54$] | .0229 | .0495 |
| persona | Hard | 11 | $-5.99$ [$-12.23$, $+0.26$] | .1147 | .3355 |
| | None | 12 | $-3.65$ [$-7.42$, $+0.13$] | .1118 | .3355 |
| | Easy | 11 | $-4.36$ [$-9.19$, $+0.48$] | .1385 | .3355 |
| BF16 difficulty | 0 | 20 | $-2.35$ [$-4.09$, $-0.62$] | .0257 | .0772 |
| | low | 11 | $-4.54$ [$-8.90$, $-0.19$] | .0860 | .1720 |
| | high | 3 | $-18.75$ [$-38.00$, $+0.50$] | .1091 | .1720 |
| task length | T1 | 9 | $-5.09$ [$-9.81$, $-0.38$] | .0755 | .1511 |
| | T2 | 8 | $-0.78$ [$-4.46$, $+2.90$] | .7270 | .7270 |
| | T3 | 17 | $-6.17$ [$-10.49$, $-1.84$] | .0189 | .0567 |
| issue type | mms | 14 | $-1.15$ [$-3.33$, $+1.04$] | .3880 | .4434 |
| | mobile data | 11 | $-9.09$ [$-14.98$, $-3.20$] | .0111 | .0333 |
| | service | 9 | $-4.40$ [$-10.32$, $+1.52$] | .2217 | .4434 |
| faults | T1 | 19 | $-6.37$ [$-9.91$, $-2.83$] | .0030 | .0091 |
| | T2 | 10 | $-1.94$ [$-7.58$, $+3.69$] | .5704 | .5704 |
| | T3 | 5 | $-2.92$ [$-5.18$, $-0.66$] | .0337 | .0674 |

Table 12: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, $n$ is the number of tasks and $G$ is $G_{\mathrm{all}}$ over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: $n$ is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-style W4), and $G$ is the level's contribution to $G_{\mathrm{all}}$, so the three levels sum to $G_{\mathrm{all}}$. Tertile cut points were fixed on telecom development tasks.

| Reference (status) | States (denominator) | Action replacing Q's (continuation) | Turn-over | Rescues / damages | Net | Paired comparison [interval; 95% unless marked]; reading |
|---|---|---|---|---|---|---|
| Q′: second W4 checkpoint (frozen before outcomes) | each design's own first divergent states; q4, q8 GPTQ-W4; 85 tasks × 6 blocks, 1020 units (% of units) | B (Q) | 16.8 | 8.0 / 8.7 | −0.7 | turnover, B minus Q′ (primary): −1.0; fixed pool [−3.9, +1.9], new tasks [−4.5, +2.7]; not resolved, equivalence not shown; net: −1.6 [−6.2, +3.0], [−6.5, +3.6]; the state sets differ |
| | | Q′ (Q) | 17.7 | 9.3 / 8.4 | +0.9 | |
| Common states (pre-registered at three blocks; extended by amendment) | 510 ordinary q8 GPTQ-W4 states, 85 tasks × 6 blocks, shared by all arms (% of states) | B (Q) | 7.1 | 4.1 / 2.9 | +1.2 | B's action beyond Q′'s, Q continues: +1.0; two-look 97% $t$ [−4.2, +6.2], bootstrap [−1.4, +3.5]; unadjusted $t$ [−3.5, +5.4], bootstrap [−1.2, +3.1]; not resolved |
| | | Q′ (Q) | 8.0 | 4.1 / 3.9 | +0.2 | |
| | | B (B; vs. QB) | 9.0 | 5.3 / 3.7 | +1.6 | B's action beyond Q′'s, B continues: +0.8; two-look 97% [−2.2, +3.8], [−2.0, +3.7]; unadjusted $t$ and bootstrap [−1.8, +3.3]; not resolved; equivalence: unadjusted 90% $t$ [−1.2, +2.8] inside ±3, two-look 94% $t$ [−1.6, +3.2] not; no equivalence claimed |
| | | Q′ (B; vs. QB) | 8.6 | 4.7 / 3.9 | +0.8 | |
| Key order (exploratory; code frozen before outcomes) | 300 q8 GPTQ-W4 first divergent states whose action is one tool call with at least two argument keys, 41 tasks (% of states) | B (Q) | 9.0 | 3.3 / 5.7 | −2.3 | turnover, reversed keys minus B: −2.3 [−8.4, +2.9]; not resolved; no equivalence margin |
| | | Q, keys reversed (Q) | 6.7 | 3.3 / 3.3 | 0.0 | |

Table 13: Reference experiments: values and paired intervals of Figure 4 (per-arm intervals: Appendices D.1 and D.3). Each row group has its own state source and denominator; comparisons are paired only within a group, and the references are never subtracted from B's turnover. Turnover, rescues, damages and net compare the replacing action with Q's action under the same continuation: against QQ when Q continues, and against Q's action followed by B (QB) when B continues. Intervals are those each protocol registers: Q′, fixed pool ($t$ over 6 block means) and new tasks (crossed); common states, $t$ over block means and task-cluster bootstrap, a reading needs both, under the two-look (Pocock) reading that the protocol designates confirmatory and, unadjusted, at the frozen levels (Appendix D.2); key order, the wider of block and crossed bootstrap.

**A. Frozen trigger and matched hindsight oracle**

| Budget | sel. | oracle r/d/net (pp) | frozen r/d/net (pp) | net ratio |
|---|---|---|---|---|
| 5% | 50 | 34/0/+34 (+3.33) | 6/2/+4 (+0.39) | 4/34 |
| 10% | 114 | 38/0/+38 (+3.73) | 7/5/+2 (+0.20) | 2/38 |
| 20% | 222 | 38/0/+38 (+3.73) | 16/11/+5 (+0.49) | 5/38 |

**A′. Frozen trigger against matched-count random selection (Figure 5)**

| Budget | sel. | net vs. random ($p$; Holm) | r/d | turnover vs. random ($p$) |
|---|---|---|---|---|
| 5% | 50 | +4 vs. +0.21 (.0345; .1035) | 6/2 | 8 vs. 3.48 (.0182) |
| 10% | 114 | +2 vs. +0.98 (.4169; .5234) | 7/5 | 12 vs. 7.52 (.0613) |
| 20% | 222 | +5 vs. +2.38 (.2617; .5234) | 16/11 | 27 vs. 14.30 (.0002) |

**B. Post hoc KL reference (requires BF16)**

| Budget | sel. | net vs. random ($p$; Holm) | r/d | turnover vs. random ($p$) |
|---|---|---|---|---|
| 5% | 72 | −1 vs. +0.45 (.8337; 1.0000) | 5/6 | 11 vs. 4.58 (.0038) |
| 10% | 126 | +1 vs. +0.82 (.5483; 1.0000) | 8/7 | 15 vs. 7.54 (.0048) |
| 20% | 257 | +3 vs. +1.59 (.3984; 1.0000) | 18/15 | 33 vs. 16.73 ($< 10^{-4}$) |

Table 14: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Net ratio: frozen net over oracle net (not a share of rescues; the frozen rule selects 6, 7 and 16 of the oracle's rescues). Random: matched-count per-cell random-trigger expectation; $p$: exact one-sided hypergeometric; Holm over the three budgets is applied to the net $p$-values only.

| SpinQuant − QuaRot | difference [fixed pool] | [new tasks] |
|---|---|---|
| KL ($\times 10^{-3}$) | −34 [−57, −11] | [−56, −11] |
| Action difference (pp) | +0.8 [−11.4, +13.0] | [−8.2, +9.4] |
| Flip given difference (pp) | +2.7 [−10.2, +15.3] | [−7.5, +11.4] |
| $D_{cc}$ (pp) | +1.2 [−3.9, +6.2] | [−4.7, +7.1] |

Table 15: Paired diagnostic on 255 identical W4A4 states; 95% intervals, $t$ over three block differences and crossed task×block bootstrap. The KL reduction is resolved; the differences in the task quantities remain unresolved. The $D_{cc}$ row is primary contrast P2.

| Cell | U1 ($n$=255) | U2B ($n$=255) | U2Q ($n$=60) |
|---|---|---|---|
| q4 GPTQ-W4 | .090 (11/12) | .082 (10/11) | .083 (4/1)* |
| q4 AWQ-style W4 | .082 (7/14) | .078 (7/13) | .083 (3/2) |
| q4 KV8 | .047 (4/8) | .039 (5/5) | .017 (0/1) |
| q8 GPTQ-W4 | .047 (8/4) | .067 (9/8) | .050 (3/0)* |
| q8 AWQ-style W4 | .086 (14/8) | .090 (15/8) | .067 (3/1) |
| q8 KV8 | .035 (5/4) | .031 (6/2) | 0 (0/0) |

Table 16: Bank U, blocks f01–f03: per-state turnover $M$ (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). $G$ lies within $\pm 0.028$ in U1 and U2B and within $\pm 0.050$ in U2Q, and every interval includes 0. Intervals are $t$ intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

| Model:sub | (i) identical | (ii) same tool structure, text | (iii) different tool or args | (iv) act vs. talk |
|---|---|---|---|---|
| q4:U1 | 399: 0 | 309: 45 (.146) | 19: 1 (.053) | 38: 10 (.263) |
| q8:U1 | 483: 0 | 202: 28 (.139) | 47: 11 (.234) | 33: 4 (.121) |
| q4:U2B | 430: 0 | 277: 44 (.159) | 24: 4 (.167) | 34: 3 (.088) |
| q8:U2B | 470: 0 | 199: 25 (.126) | 40: 9 (.225) | 56: 14 (.250) |
| q4:U2Q | 108: 0 | 63: 10 (.159)* | 5: 0 | 4: 1 (.250) |
| q8:U2Q | 129: 0 | 33: 6 (.182)* | 11: 1 (.091) | 7: 0 |
| unique rows | 1943: 0 | 1045: 150 (.144) | 143: 26 (.182) | 169: 32 (.189) |

Table 17: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. The 58 f01 flips scored in Appendix E.1 include the two duplicated U2Q rows.

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| GPTQ-W4 | 6 | 498 | $-0.8$ $[-5.2, +3.6]$ $[-5.8, +4.2]$ | $+1.4$ $[-3.0, +5.8]$ $[-4.0, +6.8]$ | $+0.0$ $[-4.2, +4.2]$ $[-6.0, +6.0]$ |
| RTN-W4 | 5 | 415 | $+0.7$ $[-2.7, +4.2]$ $[-3.6, +5.5]$ | $+1.9$ $[-3.8, +7.7]$ $[-4.6, +8.0]$ | $+2.4$ $[-2.2, +7.0]$ $[-3.9, +8.4]$ |

Table 18: Mistral-Small-3.2-24B on its fixed 83-task conditional population. pp; fixed-pool 95% $t$ intervals, then crossed task×block 95% intervals. Missing argument-JSON errors leave RTN f04 excluded, whereas model-produced invalid tool names are scored as failures under the dated amendment above. None of the three mode intervals excludes zero.

**A. Full item pool**

| Cell | items | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 788 | 6.3 | 4.3 | 10.7 | $+2.0$ $[-0.3, +4.3]$ |
| BFCL q4 AWQ-style W4 | 788 | 5.7 | 9.5 | 15.2 | $-3.8$ $[-6.5, -1.3]$ |
| BFCL q8 GPTQ-W4 | 788 | 2.9 | 2.3 | 5.2 | $+0.6$ $[-0.9, +2.3]$ |
| BFCL q8 AWQ-style W4 | 788 | 4.1 | 2.7 | 6.7 | $+1.4$ $[-0.3, +3.3]$ |
| BFCL W4 pool | | | | | $+0.1$ $[-1.0, +1.2]$ |
| ALFWorld q8 GPTQ-W4 | 134 | 9.0 | 12.7 | 21.6 | $-3.7$ $[-11.2, +4.5]$ |
| ALFWorld q8 AWQ-style W4 | 134 | 10.4 | 14.2 | 24.6 | $-3.7$ $[-11.9, +4.5]$ |
| ALFWorld W4 pool | | | | | $-3.7$ $[-9.7, +1.9]$ |
| BFCL q4 GPTQ-W3$^s$ | 591 | 6.1 | 3.6 | 9.6 | $+2.5$ $[+0.2, +5.1]$ |
| BFCL q8 GPTQ-W3$^s$ | 788 | 5.2 | 4.1 | 9.3 | $+1.1$ $[-1.0, +3.3]$ |
| BFCL q4 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| BFCL q8 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| ALFWorld q8 GPTQ-W3$^s$ | 134 | 6.7 | 3.0 | 9.7 | $+3.7$ $[-1.5, +9.7]$ |
| ALFWorld q8 KV4$^s$ | 134 | 0.0 | 0.0 | 0.0 | $+0.0$ $[+0.0, +0.0]$ |

**B. First divergent states**

| Cell | states | $G$ | $H$ | $F$ | $G - H$ | off-diag. % |
|---|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 573 | $+2.8$ $[-0.4, +5.9]$ | $+1.4$ $[-0.7, +3.5]$ | $+4.2$ $[+0.9, +7.2]$ | $+1.4$ $[-1.8, +4.5]$ | 11.7 |
| BFCL q4 AWQ-style W4 | 642 | $-4.7$ $[-7.8, -1.5]$ | $-1.1$ $[-3.6, +1.4]$ | $-1.4$ $[-4.8, +1.7]$ | $-3.6$ $[-6.7, -0.2]$ | 17.1 |
| BFCL q8 GPTQ-W4 | 533 | $+0.9$ $[-1.3, +3.3]$ | $+1.5$ $[+0.0, +3.2]$ | $+2.1$ $[-0.4, +4.6]$ | $-0.6$ $[-3.1, +2.0]$ | 4.7 |
| BFCL q8 AWQ-style W4 | 548 | $+2.0$ $[-0.4, +4.7]$ | $+1.8$ $[+0.0, +3.7]$ | $+3.6$ $[+0.9, +6.5]$ | $+0.2$ $[-2.6, +3.1]$ | 7.8 |
| BFCL W4 pool | | $+0.3$ $[-1.2, +1.8]$ | $+0.9$ $[-0.2, +2.0]$ | $+2.1$ $[+0.3, +3.9]$ | $-0.6$ $[-2.3, +1.0]$ | 10.3 |
| ALFWorld q8 GPTQ-W4 | 115 | $-4.3$ $[-13.3, +5.0]$ | $+7.0$ $[-2.6, +16.7]$ | $+9.6$ $[-0.9, +19.7]$ | $-11.3$ $[-20.9, -1.8]$ | 40.0 |
| ALFWorld q8 AWQ-style W4 | 119 | $-4.2$ $[-13.6, +4.9]$ | $+5.0$ $[-2.6, +12.2]$ | $+4.2$ $[-5.9, +13.7]$ | $-9.2$ $[-18.8, +0.0]$ | 32.8 |
| ALFWorld W4 pool | | $-4.3$ $[-11.1, +2.2]$ | $+6.0$ $[-0.5, +12.6]$ | $+6.9$ $[-2.1, +15.5]$ | $-10.3$ $[-17.7, -2.9]$ | 36.4 |
| BFCL q4 GPTQ-W3$^s$ | 540 | $+2.8$ $[+0.2, +5.5]$ | $+5.2$ $[+2.9, +7.7]$ | $+8.9$ $[+5.7, +12.2]$ | $-2.4$ $[-5.7, +0.7]$ | 13.0 |
| BFCL q8 GPTQ-W3$^s$ | 696 | $+1.3$ $[-1.1, +3.7]$ | $+1.4$ $[-0.1, +2.9]$ | $+3.4$ $[+0.9, +6.0]$ | $-0.1$ $[-2.6, +2.4]$ | 8.3 |
| BFCL q4 KV4$^s$ | 787 | $+0.3$ $[+0.0, +0.6]$ | $+2.9$ $[+1.9, +4.1]$ | $+20.3$ $[+17.6, +23.0]$ | $-2.7$ $[-3.9, -1.5]$ | 20.5 |
| BFCL q8 KV4$^s$ | 788 | $+0.3$ $[+0.0, +0.6]$ | $+1.8$ $[+1.0, +2.7]$ | $+14.5$ $[+12.1, +17.0]$ | $-1.5$ $[-2.5, -0.6]$ | 14.7 |
| ALFWorld q8 GPTQ-W3$^s$ | 130 | $+3.8$ $[-1.5, +9.8]$ | $+38.5$ $[+29.8, +46.9]$ | $+42.3$ $[+33.6, +50.8]$ | $-34.6$ $[-43.5, -25.4]$ | 30.8 |
| ALFWorld q8 KV4$^s$ | 134 | $+0.0$ $[+0.0, +0.0]$ | $+38.1$ $[+30.6, +45.5]$ | $+53.0$ $[+44.8, +61.2]$ | $-38.1$ $[-45.5, -30.6]$ | 26.9 |

Table 19: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue $r$, damage $d$, turnover $M$ (% of all items) and net $G_{\text{all}}$ (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$. Pools: equal weight per cell. $^s$Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to $H$.

| Contrast | blocks | $\Delta D_{cc}$ [fixed pool] | [new tasks] | Holm $p$ |
|---|---|---|---|---|
| P1: QuaRot − GPTQ W4 | 3 | $-0.4$ $[-10.7, +9.9]$ | $[-5.9, +5.5]$ | 1.0 |
| P2: SpinQuant − QuaRot W4A4 | 3 | $+1.2$ $[-3.9, +6.2]$ | $[-4.7, +7.1]$ | 1.0 |
| P3: KIVI-2 − KV8 | 2 | $-0.6$ $[-38.0, +36.8]$ | $[-7.1, +7.1]$ | 1.0 |

Table 20: Complete primary method-comparison family on shared states, pp; fixed-pool and new-task 95% intervals. All three comparisons remain unresolved. Admission and re-run dependencies are described in the text.

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| AWQ-style W4 | 3 | 255 | $+2.7$ [$-4.6$, $+10.1$] [$-2.4$, $+9.0$] | $+4.3$ [$-0.1$, $+8.8$] [$-1.2$, $+11.0$] | $+5.5$ [$+1.0$, $+10.0$] [$-0.8$, $+12.5$] |
| GPTQ-W4 | 3 | 255 | $+0.4$ [$-7.0$, $+7.7$] [$-4.3$, $+5.9$] | $+0.8$ [$-5.3$, $+6.9$] [$-5.9$, $+7.5$] | $+3.5$ [$-2.3$, $+9.4$] [$-3.9$, $+10.6$] |
| KIVI-16 path | 3 | 255 | $+0.0$ [$-2.9$, $+2.9$] [$-2.7$, $+2.7$] | $+1.2$ [$-6.6$, $+8.9$] [$-3.5$, $+5.9$] | $+2.4$ [$-0.6$, $+5.3$] [$-2.4$, $+7.1$] |
| KIVI-2 | 2 | 170 | $-2.9$ [$-25.4$, $+19.5$] [$-8.2$, $+2.4$] | $+2.9$ [$-4.5$, $+10.4$] [$-3.5$, $+10.0$] | $+2.9$ [$-34.4$, $+40.3$] [$-5.9$, $+10.6$] |
| KIVI-4 | 3 | 255 | $+0.0$ [$-5.1$, $+5.1$] [$-3.5$, $+4.3$] | $+2.0$ [$-1.4$, $+5.3$] [$-3.5$, $+7.5$] | $+0.8$ [$-8.1$, $+9.7$] [$-5.9$, $+7.1$] |
| KV8 | 3 | 255 | $+1.6$ [$-0.1$, $+3.3$] [$-1.2$, $+4.7$] | $+5.1$ [$+0.6$, $+9.6$] [$+0.4$, $+9.8$] | $+5.5$ [$+1.0$, $+10.0$] [$+0.0$, $+10.6$] |
| QuaRot-W16 path | 3 | 255 | $+0.0$ [$-5.1$, $+5.1$] [$-3.5$, $+3.1$] | $+1.6$ [$-14.5$, $+17.7$] [$-5.9$, $+9.4$] | $+2.4$ [$-8.2$, $+12.9$] [$-4.3$, $+9.0$] |
| QuaRot-W4 | 3 | 255 | $+0.8$ [$-5.3$, $+6.9$] [$-3.5$, $+5.9$] | $+1.6$ [$-1.8$, $+4.9$] [$-4.3$, $+7.5$] | $+4.7$ [$+1.8$, $+7.6$] [$-2.0$, $+11.4$] |
| QuaRot-W4A4 | 3 | 255 | $+1.2$ [$-6.6$, $+8.9$] [$-4.7$, $+6.3$] | $+6.3$ [$-8.4$, $+21.0$] [$-2.4$, $+14.1$] | $+9.4$ [$+1.7$, $+17.1$] [$+2.0$, $+16.5$] |
| SpinQuant-W16 path | 3 | 255 | $-0.4$ [$-2.1$, $+1.3$] [$-3.1$, $+2.4$] | $+2.0$ [$-1.4$, $+5.3$] [$-2.4$, $+6.3$] | $+2.4$ [$-0.6$, $+5.3$] [$-2.4$, $+7.1$] |
| SpinQuant-W4A4 | 3 | 255 | $+3.1$ [$-2.9$, $+9.2$] [$-2.0$, $+8.2$] | $+11.8$ [$+6.7$, $+16.8$] [$+5.1$, $+18.4$] | $+13.7$ [$+9.3$, $+18.2$] [$+5.5$, $+21.6$] |

Table 21: All q8 ordinary BF16-sourced states admitted by the panel rules, including unquantized path controls. Effects in pp; each cell: estimate with fixed-pool 95% $t$ interval, then crossed task$\times$block 95% interval. All three modes use the same admitted states within a row. These descriptive intervals are unadjusted; the primary method family is P1–P3. KIVI-2 admits only two blocks after one-time infrastructure re-runs; it is not estimable under the original strict rule. Other cells admit three blocks. No action-only interval excludes zero; several takeover or joint-upgrade intervals do.